A Weight-Free Benchmark for Multilateral Tariff Bargaining

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Quantitative models of trade negotiations divide the gains from an agreement using bargaining weights estimated from the concessions countries actually made. We ask what the same model implies when the weights are removed. On the six-country 1990 negotiating panel and calibrated primitives of Bagwell, Staiger, and Yurukoglu (2021), we replace their estimated bilateral Nash weights with a cooperative benchmark: each coalition is valued by the general-equilibrium surplus its members can attain jointly, and the surplus of the full group is divided by the Shapley (1953) value. The benchmark contains no bargaining-weight parameter. Their calibration, counterfactuals and equilibrium map reproduce; their weight estimator does not, because deal/no-deal branching in the bilateral fixed point makes its criterion a step function at the resolution of its own Jacobian — a caveat on the comparator, not on the benchmark. Relative to their estimated outcome, the benchmark favors the United States and Canada and disfavors Japan; the signs are robust across four allocation rules, the magnitudes are not, and a symmetric-weight placebo shows the gap measures the asymmetry their estimation found rather than a structural distance from the Shapley value. The benchmark lies in the transferable-utility core, but is implementable only with transfers the GATT does not provide: without them the efficient policy itself violates Australia's participation constraint. The bargained outcome falls short of a full-schedule line-by-line optimum by 79.5% of the group's surplus, of which four-fifths is agenda coverage rather than bargaining performance.

Keywords: tariff negotiations, multilateral bargaining, Shapley value, cooperative game theory, quantitative trade models.

JEL codes: F13, F17, C71, C68.

Department of Economics, Haslam College of Business, University of Tennessee, Knoxville. Email: `pli26@vols.utk.edu`. This paper builds on and replicates Bagwell, Staiger, and Yurukoglu (2021), whose replication package makes the exercise possible. All remaining errors are my own.

1. Introduction

Who gains how much from a trade agreement is the question quantitative models of trade negotiations were built to answer. The modern answer runs through bargaining weights. In the leading structural treatment, Bagwell, Staiger, and Yurukoglu (2021) — hereafter BSY — embed a Horn–Wolinsky network of bilateral Nash bargains (Horn and Wolinsky 1988) inside a calibrated general-equilibrium trade model and estimate, by GMM from observed tariff concessions, the weight that best rationalizes what negotiators agreed to. The estimated weights then carry the division of the gains. This is the right way to learn what bargaining power was; it cannot, by construction, say what the gains would have been divided into absent that bargaining power, because the weights are recovered from the outcome one might want to evaluate. The literature has no weight-free yardstick against which an observed agreement can be read.

This paper builds one and takes it to the same economy. We hold BSY's calibrated primitives, their six-country 1990 negotiating panel (United States, European Union, Japan, Canada, Australia, South Korea), and their policy space fixed, and change only how the surplus is generated and divided. Every coalition S of negotiating countries is assigned a value $v(S)$: the general-equilibrium dollar surplus its members attain by choosing their tariff schedules jointly to maximize their own combined real income, with non-members held at their 1990 applied tariffs. The grand coalition's surplus is then divided by the Shapley (1953) value, the unique allocation satisfying efficiency, symmetry, the dummy property, and additivity (Hart and Mas-Colell 1989). Neither step contains a bargaining weight. Because the optimal policy that generates $v(S)$ is computed before, and independently of, the rule used to divide the resulting surplus, the benchmark is a normative object: it says what an efficient, axiomatically fair division would have looked like, not what bargaining power was.

The Shapley value is the natural rule here for three reasons. Restricted to two players it coincides with the symmetric Nash bargaining solution (Proposition 1), so it extends BSY's own bilateral benchmark to $N > 2$ rather than replacing it. It pays each country its expected marginal contribution to a randomly formed coalition, which is the formal counterpart of the reciprocity intuition in Bagwell and Staiger (2002) that negotiating leverage tracks the externality a country's tariffs impose on partners. And it is unique under four axioms, where competing concepts are either set-valued or lack the closed-form computability a model that must be re-solved at every coalition requires. Agnew (2023, 2025) propose the Shapley value for cooperative trade gains in a stylized three-country example; we derive $v(S)$ from a structural general-equilibrium model rather than stipulating it, enumerate all 63 coalitions of BSY's own panel exactly, and, as a robustness check, repeat the exercise on the 31-country, 40-sector economy of Caliendo and Parro (2015).

The comparator, and what does not reproduce. The exercise requires BSY's economy to be reproduced first. That reproduction is a precondition for what follows rather than a contribution of it; we report it briefly, and at all, because one of its four layers does not go through. Their calibrated arrays load to a maximum absolute difference of 2.5×10^{-14}, their published counterfactual reproduces to 0.024 percentage points, and their Nash-in-Nash equilibrium map reproduces from a cold start to 0.40 percentage points of mean tariff deviation. Their weight estimator does not reproduce: our re-estimation diverges. The mechanism is identifiable, and it bears directly on how much weight the estimated comparator can carry. Each bilateral pair in the Nash-in-Nash fixed point takes a discrete deal/no-deal branch on every tariff line, so two nominally identical re-solves of the same weight vector can land on different branches; the resulting jumps in the GMM criterion are of the same order as the finite-difference step used to build its Jacobian. At that resolution the criterion is better described as a path-dependent step function than as a smooth objective — which is also why BSY's own estimation required a coarse minimum differencing step. Nothing downstream depends on the failed re-estimation: the benchmark carries no weight at all, and every weight-dependent comparator we report is evaluated at BSY's own published estimate $\hat{\zeta}$ and their own cached equilibrium.

Result 1: the benchmark and the bargain disagree about Japan. We compare the two allocations through a *fairness gap*, $\Gamma_i = \phi_i(v)/v(\mathcal{N}) - s_i^{HW}(\hat{\zeta})$, the difference between country i's Shapley share of the cooperative surplus and its share of BSY's estimated bargained surplus. Japan is the only negative gap and the largest in magnitude: the estimated bargain awards Japan 55.5% of the surplus where the benchmark assigns 42.9%. The signs of the United States (+3.3pp), Japan (−12.5pp), and Canada (+2.1pp) gaps are robust across four efficient, individually rational allocation rules (Shapley, equal split, 1990 revenue share, 1990 income share); the magnitudes are not — the United States gap ranges from +3.3 to +23.2pp across rules — and three countries change sign. Two further checks discipline the reading. A symmetric-weight placebo, $\zeta = 0.5$, moves the United States gap to −0.06pp and flips four of six signs: the gap at $\hat{\zeta}$ measures the size of the asymmetry BSY's estimation found, not a structural distance between cooperative benchmarks and bargaining outcomes. And 97% of the cross-country variation in the benchmark is already present in the efficient policy's own physical incidence, which the Shapley value compresses by about a quarter; the United States gap sits inside the part a one-parameter shrinkage of that incidence already explains.

Result 2: the fair allocation is self-enforcing only with transfers. The game is not superadditive on this economy — adding the United States to {EU, Canada} lowers the coalition's cooperative value, and 20 of 301 disjoint-pair tests fail — so core membership cannot be inferred from convexity and must be checked. It holds: all 62 proper-coalition constraints are satisfied, with a minimum slack of 2.59% of the grand-coalition surplus. But this is a property of the *transferable-utility* game. Without a transfer instrument, the efficient policy itself is blocked: Australia is worse off at the surplus-maximizing tariff vector than at its own 1990 status quo, and moving from the efficient policy's incidence to the Shapley allocation requires a budget-balanced gross transfer of 13.5% of the group's surplus. The GATT/WTO is by design a no-transfer institution, operating through issue linkage and reciprocal concessions rather than side payments, so this is a first-order implementability caveat on the benchmark, not a technical footnote.

Result 3: the shortfall is mostly agenda, not bargaining. BSY's estimated bargain delivers $\Sigma_k g_k^{HW} = 0.3124$ against a grand-coalition value of $v(\mathcal{N}) = 0.7202$ in their own income units — a level gap of 57% of the group's surplus that invites, and does not support, the reading that bilateral bargaining left half the gains on the table. Attributing it requires holding fixed what varies. We therefore re-solve the same optimization on BSY's own line-by-line instrument, first restricted to the 151 tariff lines their estimation sample negotiates and then extended to the full 294-line schedule, and decompose the gap against the full-coverage optimum into three matched-instrument terms: extending coverage beyond BSY's own negotiating set is worth 62.2% of $v(\mathcal{N})$; imposing individual rationality at the 1990 threat point costs 1.2%; and the residual within-agenda bargaining shortfall is 16.1%. Widening what is negotiable is worth 3.88 times more than bargaining harder over what already is. The within-agenda shortfall is not an artifact of the threat point: replacing 1990 with

BSY's own realized outcome as every member's floor leaves 85% of it intact as a no-transfer-feasible weak Pareto improvement, though four of six members are exactly indifferent at that point and the entire net gain accrues to Japan and Canada. The three-way split is exact by construction of the nested programs, but the reading of its last term is not clean: cut depth on the shared lines and BSY's seven-pair bilateral architecture remain jointly embedded in it, and a dedicated attempt to separate them returned a boundary result rather than a verdict (Section 4.4).

Result 4: an external check, and its placebo. The benchmark's own optimal line-by-line cuts correlate positively with the tariff changes actually observed on those lines between 1990 and 2000 (Spearman $\rho = +0.64$), the paper's one comparison against data outside the model. It does not survive its placebo. A model-free rule that cuts every line by the same proportion scores higher ($\rho = +0.71$), and a paired bootstrap resolves the difference against the benchmark. On the extensive margin the contrast with BSY's own fitted schedule is nonetheless sharp: their cached equilibrium leaves 86 of the 151 negotiated lines (57%) at their solver's no-cut tolerance, while the efficient frontier cuts 118 of them (78%). A rank comparison favors the benchmark on the pooled sample ($\rho = +0.32$ for BSY) but not on the 65 lines where both instruments actually move, so we report the comparison as unresolved at the coverage-matched caliber. Neither schedule carries information about the observed cuts beyond simple persistence of the 1990 tariff once rank is partialled on it.

What we do not claim. Following Manski (1995), identification requires that distinct parameter values generate distinct distributions of observed outcomes. Our framework does not meet that bar, and the reason is sharper than the generic caveat. The policy that generates $v(S)$ is chosen independently of the rule that divides the resulting surplus, so the Shapley value, the nucleolus, the Kalai–Smorodinsky solution and the γ-core of Chander (2007) all imply exactly the same tariffs at every coalition. No tariff moment can discriminate among them; only observed international transfers could, and the GATT does not generate them. We therefore do not claim to identify bargaining power, to identify the allocation rule, or to read the Shapley ranking as evidence about strategic position. The threat point compounds this: the 1990 applied schedule is itself the outcome of four decades of prior GATT bargaining (Limão 2007), so it is not exogenous to the object we study. The benchmark is a computed normative object under stated assumptions, and we report it as one.

Related literature. The paper is closest to the quantitative bargaining literature that embeds Nash-in-Nash solutions in structural models (Bagwell, Staiger, and Yurukoglu 2021, 2020; Collard-Wexler, Gowrisankaran, and Lee 2019), and takes from it both the economy and the comparator. It departs from that literature in replacing the estimated weight with an axiomatic allocation, which connects it to the cooperative-game treatment of trade agreements (Agnew 2023, 2025; Rogna 2016) and to the coalition-formation literature on externalities and the γ-core (Chander 2007; Ray and Vohra 2024). The terms-of-trade foundation for reciprocity we operationalize is Bagwell and Staiger (2002, 1999); the general-equilibrium machinery, including exact hat-algebra and the welfare aggregation we use, is Eaton and Kortum (2002), Caliendo and Parro (2015), and Arkolakis, Costinot, and Rodriguez-Clare (2012). The computational strategy — treating the trade equilibrium as constraints of the tariff-optimization problem rather than as an inner loop — follows the MPEC formulation of Luo, Pang, and Ralph (1996) and, in the trade context, Ossa (2014). Relative to all of these, our contribution is to make an axiomatic division operational inside a calibrated general-equilibrium model, on the same panel and primitives an estimated bargaining model was fit to, and to report what the comparison does and does not support.

Section 2 sets out the cooperative benchmark. Section 3 describes BSY's calibration and the second economy. Section 4 reports the results above on BSY's own panel; Section 5 reports the second economy and the sensitivity of every headline number. Section 6 concludes. Proofs are in Section A; supporting tables, the full second-economy results, and a map from every number in the text to the artifact that produced it are in the remaining appendices.

2. A Cooperative Benchmark for Tariff Bargaining

The benchmark has three components: a general-equilibrium trade model that maps tariffs into welfare, a characteristic function that assigns each group of countries the surplus it can generate, and an allocation rule that divides the surplus of the full group. The first is taken from the literature unchanged. The second and third are where this paper's framework departs from an estimated bargaining model, and they are the subject of Sections 2.2 and 2.3. Section 2.4 states the conditions under which the resulting allocation can be compared to an estimated one.

2.1. Trade equilibrium

We use the multi-sector Eaton–Kortum structure of Caliendo and Parro (2015). The world has N countries, indexed n, i, h, and J sectors, indexed j, k. Each country-sector produces a continuum of varieties with Fréchet productivity draws of shape θ^j, using labor and intermediates from all sectors according to an input–output matrix $\gamma_n^{k,j}$ with labor share γ_n^j. Shipping sector-j goods from i to n costs an iceberg factor $d_{ni}^j \geq 1$ and an ad-valorem tariff $\tau_{ni}^j \geq 0$, so delivered cost scales by $\kappa_{ni}^j \equiv (1 + \tau_{ni}^j) d_{ni}^j$. Final demand is Cobb–Douglas with sector shares α_n^j.

We solve counterfactuals in exact hat-algebra. Writing $\widehat{x} \equiv x'/x$ for the ratio of a counterfactual to its base-year value, and holding technology and iceberg costs fixed, the equilibrium in changes is

$$\widehat{c}_n^j = \widehat{w}_n^{\gamma_n^j} \prod_k (\widehat{P}_n^k)^{\gamma_n^{k,j}}, \tag{1}$$

$$\widehat{P}_n^j = \left[\textstyle\sum_i \pi_{ni}^j (\widehat{\kappa}_{ni}^j \widehat{c}_i^j)^{-\theta^j}\right]^{-1/\theta^j}, \tag{2}$$

$$\widehat{\pi}_{ni}^j = (\widehat{c}_i^j \widehat{\kappa}_{ni}^j / \widehat{P}_n^j)^{-\theta^j}, \tag{3}$$

$$X_n^{j\prime} = \textstyle\sum_k \gamma_n^{j,k} \sum_i \frac{\pi_{in}^{k\prime} X_i^{k\prime}}{1+\tau_{in}^{k\prime}} + \alpha_n^j I_n', \tag{4}$$

$$\textstyle\sum_{j,i} \frac{\pi_{ni}^{j\prime} X_n^{j\prime}}{1+\tau_{ni}^{j\prime}} - D_n = \sum_{j,i} \frac{\pi_{in}^{j\prime} X_i^{j\prime}}{1+\tau_{in}^{j\prime}}, \tag{5}$$

an exactly determined system in $(\widehat{w}_n, \widehat{P}_n^j, \widehat{c}_n^j, \widehat{\pi}_{ni}^j, X_n^{j\prime})$ once a numeraire is fixed. The unobserved fundamentals cancel between x' and x, so only base-year trade shares $\{\pi_{ni}^j, X_n^j\}$, the elasticities θ^j, and the tariff change enter (Eaton and Kortum 2002; Caliendo and Parro 2015). Real income is $W_n = I_n / \prod_j (P_n^j)^{\alpha_n^j}$, so

$$\widehat{W}_n = \widehat{I}_n \Big/ \textstyle\prod_j (\widehat{P}_n^j)^{\alpha_n^j}, \tag{6}$$

which by Arkolakis, Costinot, and Rodriguez-Clare (2012) captures the full general-equilibrium welfare effect of the tariff change. This machinery is entirely standard; what follows is not.

2.2. The characteristic function

Policy space. Each importer n controls a single scalar that scales its entire applied schedule proportionally,

$$\tau_{ni}^{j,\text{new}} = s_n\, \tau_{ni}^{j,0}, \qquad s_n \in [0,1]\ \forall n \in S, \qquad s_n = 1\ \forall n \notin S. \tag{7}$$

The restriction to a scalar per country is what makes exact enumeration over 2^N coalitions feasible: it reduces each coalition's decision space from $\sim N^2 J$ to $|S|$. It is a tractability device and not a description of a negotiating instrument — actual rounds bargain line by line, and Section 4 re-solves the central comparisons on BSY's own line-by-line instrument precisely because the two are not interchangeable. The upper bound $s_n \leq 1$ is not a modeling convenience either: it is the GATT bound-rate constraint, and it binds. Four of the six countries' unconstrained unilateral optimum is to *raise* tariffs, in line with the optimal-tariff magnitudes in Ossa (2014).

Value of a coalition. For a coalition S, let I_n^{base} denote country n's base-year income. The characteristic function is the unconstrained maximum of the members' combined real-income gain,

$$v(S) = \max_{s_S \in [0,1]^{|S|}} \sum_{n \in S} I_n^{\text{base}} \big(\widehat{W}_n(s_S, \mathbf{1}_{-S}) - 1\big), \qquad v(\emptyset) = 0, \tag{8}$$

where $\widehat{W}_n$ solves Eqs. (1) to (5) at the coalition's policy and non-members are held at their base-year schedule.

Three features of Eq. (8) matter, and each is a deliberate departure from the bargaining formulation it replaces. First, no bargaining weight appears anywhere: the maximization is over a compact box with a linear objective, so it has a solution by Weierstrass, and any two maximizers attain the same value. Second, no individual-rationality constraint appears inside the program. The economic content of "no member does worse than its outside option" is not discarded; it is relocated to the allocation layer, where we verify $\phi_n(v) \geq v(\{n\})$ directly (Section 2.3). Third, non-members are passive: they keep their base-year tariffs and are not protected from the coalition's action. This is a model of plurilateral liberalization with free-riding — a member that cuts its applied MFN tariff cuts it on non-members too, who receive the market access without reciprocating — rather than of a discriminatory bloc (Chander 2007). Non-negativity is then constructive rather than assumed: $s_S = \mathbf{1}$ is feasible and yields the status quo, so $v(S) \geq 0$ for every S.

The alternative we do not use is a weighted Nash product over the coalition's members. It fails on its own terms in this setting: the Nash objective and the dollar payoff are different functions, so its argmax does not define the value of any single program; on coalitions whose individually rational set contains no strict Pareto improvement the product is identically zero and the numerical argmax reports the smallest dollar element of that face; and it puts a bargaining weight inside the characteristic function, which is the object we are trying to build without one.

Utilitarian weights, and what is and is not weight-free. Removing the bargaining weight does not make policy selection distribution-free. The objective in Eq. (8) maximizes an unweighted dollar sum, which is a Negishi-type weighting of utilities by the inverse marginal utility of income and is not neutral across policies that redistribute differently across countries. What is weight-free is narrower and is the claim we make: the *division* of the resulting surplus is chosen after, and independently of, the policy, so alternative allocation rules operate on an identical $v(\cdot)$ and imply identical tariffs. Bagwell, Staiger, and Yurukoglu (2021) report a world-income-maximizing row among their own comparison benchmarks, so the objective itself is not novel here; the allocation layer is.

Cooperative normalization. A singleton value $v(\{n\})$ is the gain country n obtains by optimizing its own tariff alone, which is an outside-option effect on its own payoff and not cooperative surplus. We therefore zero-normalize,

$$w(S) \equiv v(S) - \sum_{n \in S} v(\{n\}), \qquad w(\{n\}) = 0 \text{ by construction}, \tag{9}$$

and decompose $v(N) = \sum_n v(\{n\}) + w(N)$. By linearity of the Shapley value the two games' allocations decompose exactly, $\phi_n(v) = v(\{n\}) + \phi_n(w)$. We report the cooperative allocation $\phi(w)$ as the paper's primary normative object and the raw allocation $\phi(v)$ when comparing to an estimated bargaining outcome, for the reason given in Section 2.4: the comparator's gains are measured from the same 1990 status quo with nothing netted out, and the two objects must share a threat point to be differenced. On the six-country panel the distinction is small — four of six countries have $v(\{n\}) = 0$, so $\phi_n(v) = \phi_n(w)$ for them — and on the second economy of Section 5 it is decisive.

Single-step formulation. Combining Eqs. (7) and (8) with the equilibrium conditions gives a mathematical program with equilibrium constraints (Luo, Pang, and Ralph 1996), in which the equilibrium variables are decision variables rather than the output of an inner loop:

$$\begin{aligned} \max_{\{s_n\}_{n \in S},\, \widehat{w}, \widehat{P}, \widehat{c}, \widehat{\pi}, X'} \quad & \sum_{i \in S} I_i^{\text{base}} \big(\widehat{W}_i - 1\big) \\ \text{s.t.} \quad & \text{Eqs. (1) to (5)}, \\ & \widehat{\kappa}_{ni}^{j}(s_n) = (1 + s_n \tau_{ni}^{j,0})/(1 + \tau_{ni}^{j,0}), \\ & s_n \in [0,1] \ \forall n \in S, \qquad s_n = 1 \ \forall n \notin S. \end{aligned} \tag{10}$$

One trade-balance constraint is redundant by Walras' law and is dropped. Our implementation solves Eq. (10) by continuous quasi-Newton optimization with exact implicit-function-theorem gradients through the equilibrium, from multiple starts. Grid or vertex enumeration would be invalid here: the grand-coalition optimum on BSY's panel is interior on five of six coordinates and exceeds the best vertex by 17.4%.

2.3. Allocation and stability

DEFINITION 1 (Shapley value). *For a transferable-utility game $v : 2^N \to \mathbb{R}$,*

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(N - |S| - 1)!}{N!} \big(v(S \cup \{i\}) - v(S)\big). \tag{11}$$

ϕ is the unique rule satisfying efficiency, symmetry, the dummy property, and additivity (Shapley 1953; Hart and Mas-Colell 1989). Two properties connect it to the bilateral benchmark it generalizes.

PROPOSITION 1 (Two players). *Let $N = \{i, j\}$. On the cooperative game w, $\phi_i(w) = \phi_j(w) = \frac{1}{2}w(N)$, which is the symmetric Nash bargaining solution of the two-player transferable-utility game with disagreement point $d_n = v(\{n\})$. On the raw game v, $\phi_i(v) = v(\{i\}) + \frac{1}{2}w(N)$, which coincides with the symmetric Nash bargain measured from the 1990 status quo — the disagreement point of Bagwell, Staiger, and Yurukoglu (2021) — if and only if $v(\{i\}) = v(\{j\})$.*

Proofs of this and all subsequent formal statements are in Section A. Proposition 1 places the benchmark relative to BSY exactly: our framework nests their bilateral bargain when their estimated weight is symmetric, and does not nest the asymmetric case, which would require the weighted Shapley value of Kalai and Samet (1987) with exogenous weights. The value added by the Shapley value over a bilateral Nash bargain therefore appears only at $N \geq 3$, through cross-coalition heterogeneity in $v(S)$.

Whether the resulting allocation is also stable is a separate question, and on this economy it cannot be answered by the usual shortcut. Shapley (1971) shows that convexity implies the Shapley value lies in the core; here the game is not even superadditive. Adding the United States to {EU, Canada} lowers the coalition's cooperative value, 20 of 301 disjoint-pair tests fail, and single-player monotonicity of w fails 17 times in 192 additions (Section 4.2). We therefore verify the two properties that matter directly rather than deriving them: individual rationality of the imputation, $\phi_n(v) \geq v(\{n\})$, and membership of ϕ in the core. Both hold on both economies we study, with cushions reported in Sections 4 and 5. One implication is worth stating because it is free: since ϕ is in the core and the Shapley value is efficient, no partition of the players can attain more in the projected game than the grand coalition, which is why we continue to report $v(N)$ as the headline value despite the superadditivity failure.

2.4. Comparing a benchmark to an estimate

The benchmark is normative; BSY's Horn–Wolinsky equilibrium is an estimate. Differencing them is informative only if the two sides are constructed on the same terms, and most of the work in this subsection is in stating those terms.

DEFINITION 2 (Fairness gap). *Let $g_i(\zeta)$ be country i's dollar gain in BSY's estimated bargaining equilibrium at weight vector ζ, and $s_i^{HW}(\zeta) = g_i(\zeta)/\sum_k g_k(\zeta)$ its share of that equilibrium's total. The raw and cooperative fairness gaps are*

$$\Gamma_i^{\text{raw}}(\zeta) = \frac{\phi_i(v)}{v(N)} - s_i^{HW}(\zeta), \tag{12}$$

$$\Gamma_i^{\text{coop}}(\zeta) = \frac{\phi_i(w)}{w(N)} - s_i^{HW}(\zeta). \tag{13}$$

Γ^{raw} is primary. BSY's gains are measured from the 1990 applied schedule with nothing subtracted, and $\phi_i(v)$ is measured from the same point; $\phi_i(w)$ has already netted out each country's unilateral gain and therefore implicitly uses a different disagreement point. Γ^{coop} is reported as a secondary object, not as an alternative headline.

Comparability conditions. Any comparison of this kind must hold five things fixed, and we state them once here so that every instance in Section 4 inherits them. *(i) Threat point*: both sides are measured from the 1990 applied tariff schedule, which is why Γ^{raw} and not Γ^{coop} is primary. *(ii) Normalization*: each side is normalized on its own total, and the difference between the totals is carried explicitly by the level term of Eq. (14) below rather than silently absorbed into the gap. *(iii) Player set*: all six negotiating countries, including Canada, which appears in none of BSY's estimated pairs but has a non-zero general-equilibrium gain and is therefore not a degenerate coordinate. *(iv) Policy space*: both sides are cuts-only, $s \in [0, 1]$. This matches BSY's own solver, whose upper bound is the 1990 applied rate; their lower bound permits small negative tariffs where ours imposes zero, so our feasible set is a strict subset and our $v(S)$ is, in that respect, a conservative bound. *(v) Transferability*: BSY's equilibrium is non-transferable — their replication code contains a complete transfer block that is hard-disabled — while the Shapley value is a transferable-utility concept. The two regimes cannot be reconciled by any choice of convention, so we do not attempt to; we measure the wedge instead, by re-solving the same program under member-by-member participation constraints (Section 4.4), and report it as a separate term.

Two further conditions bind when the comparison is made at the level of tariff lines rather than countries, and both are violated by the naive version of the comparison. *Coverage*: BSY's negotiating set is the 151 reciprocal principal-supplier lines their estimation sample constructs, which excludes agriculture and tobacco; our scalar lever moves a country's whole schedule. *Instrument granularity*: one scalar per country is strictly coarser than one number per line, so a scalar-lever value understates what the same coverage can deliver. A subtraction that crosses either boundary is a bound, not a measurement, and Section 4.4 accordingly re-solves both sides on the same instrument before differencing them.

A decomposition, not a summary statistic. The two gaps satisfy an exact dollar identity that separates the distributional comparison from the level comparison:

$$\phi_i(v) - g_i(\zeta) = v(N)\,\Gamma_i^{\text{raw}}(\zeta) + s_i^{HW}(\zeta)\Big[v(N) - \textstyle\sum_k g_k(\zeta)\Big]. \tag{14}$$

The first term is a claim about division; the second is a claim about how much surplus the bargain generated in total. Reporting a fairness gap without Eq. (14) risks presenting an efficiency comparison as a distributional one, which on this economy would be a large error: the level term is 56.6% of $v(N)$ and dwarfs every Γ_i (Section 4.2).

Definition 3 (Shapley-consistent weights). *The set of bargaining weights that reproduce the benchmark exactly is $Z_0^* = \{\zeta \in \mathcal{Z} : L(\zeta) = 0\}$, where $L(\zeta) = \|s^{HW}(\zeta) - \phi(v)/v(N)\|_2^2$ and $\mathcal{Z} = [0.005, 0.995]^K$ is BSY's own optimizer box over its $K = 7$ estimated pairs. Its numerical-tolerance neighborhood is $Z_\delta^* = \{\zeta : L(\zeta) \le \delta\}$.*

Z_δ^* is a solver-precision object, not a confidence set: it has no sampling error of its own, and we never report it as one. Whether it is non-empty is a question about the estimated model, and Section 4.2 reports what our search of it could and could not establish.

Identification scope. Of the 2^N coalitional values, only $v(N)$ has a plausible data analog, and that analog is itself a calibration target. The remaining $2^N - 2$ values are model output. More sharply, and unlike the generic partial-identification caveat: because the policy generating $v(S)$ is selected independently of the allocation rule, no allocation rule — ours or any competitor — implies a different tariff at any coalition. The choice of Shapley is therefore not merely unidentified from tariff data; it is structurally decoupled from tariff predictions altogether. Every comparison in this paper is conditional on that scope, on the solver-selected equilibrium branch discussed in Section 4.2, and on the maintained assumptions collected in Section A.

3. Data and Calibration

3.1. BSY's 1990 negotiating panel

Our primary data are the calibrated primitives of Bagwell, Staiger, and Yurukoglu (2021), used unchanged. Their world economy has 11 regions and 67 sectors for 1990. Six of the regions — the United States, the European Union, Japan, Canada, Australia and South Korea — are the negotiating countries and are the players of our cooperative game. The remaining five are "not elsewhere specified" composites that fill out the world economy; they are held fixed at their 1990 data throughout and never enter a coalition, for the same reason we exclude the rest-of-world residual from the second economy's player set (Section B.1): a composite is an accounting aggregate, and admitting it as a player would inflate coalition surplus with no strategic content. All 63 non-empty coalitions of the six players are solved exactly.

The primitives comprise bilateral iceberg trade costs, Fréchet trade elasticities, expenditure shares, sectoral input–output linkages, factor endowments, and the 1990 applied MFN tariff schedule that anchors the threat point. We load BSY's original inputs directly and reproduce every array their own counterfactual script constructs. Against a ground truth exported from that script, all 16 arrays and 3 scalars agree to a maximum absolute difference of 2.5×10^{-14}, inside the pre-registered acceptance threshold of 10^{-12}. Values are reported in BSY's own income units, and trade deficits are set to zero in every counterfactual, so that welfare changes are not driven by an exogenous transfer; all headline magnitudes are ratios to $v(\mathcal{N})$ and therefore unit-free.

Table 1 summarizes the panel. Two features drive everything that follows. Applied tariffs are low and fairly uniform — from 4.8% in the United States to 12.3% in South Korea — so the cuts-only policy space gives every country a narrow lever, and the cooperative surplus is correspondingly a large fraction of the total: only Japan and Canada gain anything from optimizing alone, so $\sum_n v(\{n\})$ is 6% of $v(\mathcal{N})$ and the raw and cooperative allocations coincide for four of the six players. Economic size, by contrast, is very unequal, and it is size rather than tariff level that drives marginal contributions here.

Table 1: The six negotiating countries in BSY's 1990 panel.

Country	Applied tariff 1990 (%)	Income share (%)	Tariff revenue share (%)	Unilateral value $v(\{n\})$
United States	4.76	33.25	31.01	0.000
European Union	7.91	36.62	36.67	0.000
Japan	9.18	23.35	16.02	0.037
Canada	7.68	2.80	3.60	0.006
Australia	9.72	1.43	4.62	0.000
South Korea	12.26	2.55	8.08	0.000

Applied tariffs are unweighted means of the 1990 applied MFN rate over the 49 traded sectors. Shares are of the six-country total. $v(\{n\})$ is the value of the singleton coalition in BSY's income units, for comparison with $v(\mathcal{N}) = 0.720$; a zero means the country's own 1990 schedule is already at or below its unilateral optimum within the cuts-only policy space.

The comparator. Every weight-dependent quantity we report — $s_i^{HW}(\hat{\zeta})$, the fairness gaps, and the level term of Eq. (14) — is evaluated at BSY's own cached Horn–Wolinsky equilibrium at their published estimate $\hat{\zeta}$, not at an equilibrium we re-solve. This is a deliberate choice with a disclosed cost. A cold-started re-solve at the identical $\hat{\zeta}$, using the same pairwise bargaining code, reproduces their equilibrium tariff map to 0.40 percentage points of mean deviation but disagrees on the discrete deal/no-deal outcome for 26 of the 151 negotiated lines: their cached solution records a deal on every line, while ours lands 26 lines at the no-deal corner. Using their cached solution keeps the comparator exactly the published bargaining outcome rather than a re-derived approximation of it, and keeps this branch-sensitivity out of every gap we report. It does not make the sensitivity disappear, and Section 4.2 reports how much the headline numbers move on an alternative branch at the same $\hat{\zeta}$.

The negotiating set. BSY's estimation sample is not the whole tariff schedule. It consists of the 151 reciprocal principal-supplier lines — out of 294 country-sector lines in the six-country panel — on which their bilateral bargaining protocol operates, and it excludes agriculture and tobacco by construction. That exclusion is a property of the estimation sample rather than of the Uruguay Round, and it matters for how the results in Section 4.4 should be read. It is not, however, an omitted liberalization channel that our lever could simply restore: the Round brought agriculture under effective discipline through tariffication of non-tariff barriers under a separate formula (Ingco 1996; Hoda 2001), not through the principal-supplier channel BSY estimate, and measured applied agricultural protection in this sample *rose* on average over 1990–2000 (pooled median +1.45pp, 32 of 54 country-sector cells rising), which is what a switch from invisible non-tariff barriers to explicit bound tariff-equivalents looks like in an applied-tariff series.

3.2. A second economy

As a check that the results are not an artifact of one six-country panel, we repeat the exercise on the calibrated 1993 dataset of Caliendo and Parro (2015): 30 countries plus a rest-of-world residual, 40 sectors, with bilateral applied MFN tariffs from UNCTAD TRAINS restricted to manufacturing. Excluding non-tariff barriers makes the welfare gains conservative for jurisdictions whose true protection exceeds applied tariffs, which is also BSY's convention. The rest-of-world residual is an accounting aggregate of roughly 130 countries and is never a player; Section B.1 documents its construction and the sensitivity of the results to it.

Because exact enumeration scales as 2^N, we report a nested sequence of coalitions chosen for heterogeneity in size and protection: $N = 3$ is NAFTA (United States, Mexico, Canada), and $N = 4, \ldots, 7$ add China, India, Brazil and Korea in turn. This economy is structurally different from the primary panel in exactly the way that makes it a useful check: applied tariffs vary by an order of magnitude across players — 3.4% in the United States against 34.5% in India — so unilateral gains are large for some members and zero for others, and the distinction between the raw and cooperative allocations, which is nearly immaterial on BSY's panel, becomes the central fact. Section 5 reports the comparison and Section B the full results.

4. Results

This section applies the benchmark of Section 2 to Bagwell, Staiger, and Yurukoglu (2021)'s own six-country economy. We first establish what of their analysis reproduces (Section 4.1), then report the benchmark allocation and its stability (Section 4.2), the fairness gap and what it does and does not measure (Section 4.3), the efficiency comparison and its decomposition (Section 4.4), and a comparison against tariff changes observed outside the model (Section 4.5).

4.1. The comparator: what reproduces, and what does not

Reproducing BSY's economy is a precondition for the benchmark, not a result of it; we record it here because the comparator every later section is measured against is theirs, and because one of its four layers does not reproduce. Three of the four do. Their calibrated primitives load to a maximum absolute difference of 2.5×10^{-14} across 16 arrays and 3 scalars. Their published counterfactual — the welfare effects of the negotiated tariff changes — reproduces to a maximum absolute deviation of 0.024 percentage points. Their Nash-in-Nash equilibrium map reproduces from a cold start, plus a search over the feasibility box, to 0.40 percentage points of mean tariff deviation at their own point estimate $\hat{\zeta}$, though with the branch disagreement documented in Section 3.1: 26 of 151 negotiated lines settle at the no-deal corner in our re-solve and at a deal in their cached solution.

The fourth layer, their GMM estimator of the bargaining weights, does not reproduce. Our re-estimation diverges, with the Levenberg–Marquardt damping parameter running from 7.4×10^{12} to 2.9×10^{18} while the finite-difference Jacobian degenerates. This is not a coding accident, and the diagnosis concerns the estimated model rather than our implementation. Each bilateral pair in the Nash-in-Nash fixed point reaches a deal on a line only if its minimum individual gain clears a tolerance, and reverts to the 1990 baseline otherwise, so the equilibrium is a step function of the weight vector. Two independent re-solves of the identical symmetric point $\zeta = 0.5$ return criterion values that differ by 4.1×10^{-4} — direct evidence that at least one line's branch is decided differently across nominally identical solves — and an isolated, fully converged re-run of that same point lands on a branch 5.08 percentage points away from the cached comparator. The differencing step used to assemble the Jacobian, chosen to match BSY's own minimum step, is of the same order as this branch noise. At that resolution the criterion is better described as a path-dependent step function than as a smooth objective, which is also the most likely reason BSY's own estimation required a coarse minimum step to begin with.

We report this as a disclosed negative result. It does not propagate: the characteristic function Eq. (8) contains no bargaining weight, and every weight-dependent object below is evaluated at BSY's own published $\hat{\zeta}$ and their own cached equilibrium. It does, however, motivate the placebo and branch checks in Section 4.3, since a comparator recovered from a fragile estimation problem deserves them.

4.2. The benchmark allocation and its stability

Solving Eq. (10) for all 63 coalitions gives a grand-coalition value of $v(\mathcal{N}) = 0.720$ in BSY's income units, of which 0.677 — 94.05% — is genuinely cooperative surplus in the sense of Eq. (9). The optimal policy is interior and strongly asymmetric, $s^* = (0.53, 0.47, 0.03, 0.33, 0.11, 0.00)$ for the United States, the European Union, Japan, Canada, Australia and South Korea, and exceeds the best corner of the policy box by 17.4%; South Korea, whose applied tariff is the highest in the panel, is asked to go to free trade while Japan, whose tariff is close behind, cuts almost as deeply. Grid or vertex methods would have missed this point entirely.

Table 2 reports the resulting allocation next to BSY's estimated bargaining outcome. The two disagree about Japan, in one direction and by a wide margin: the estimated bargain awards Japan 55.5% of the surplus where the benchmark assigns 42.9%. Every other member is under-compensated relative to the benchmark, the United States by 3.3 percentage points and Canada by 2.1.

Table 2: The cooperative benchmark and BSY's estimated bargaining outcome.

Country	Shapley benchmark $\phi_i(v)$	share (%)	Bargained share (%)	Efficient policy incidence (%)	Fairness gap (pp)	Transfer T_i
United States	0.096	13.33	10.04	15.82	+3.29	−0.018
European Union	0.180	25.02	20.71	29.15	+4.30	−0.030
Japan	0.309	42.93	55.47	49.82	−12.54	−0.050
South Korea	0.090	12.46	10.33	6.46	+2.14	+0.043
Canada	0.024	3.36	1.24	2.84	+2.12	+0.004
Australia	0.021	2.90	2.21	−4.08	+0.69	+0.050

$v(\mathcal{N}) = 0.720163$, $w(\mathcal{N}) = 0.677338$, $\sum_n v(\{n\}) = 0.042825$. "Bargained share" is $s_i^{HW}(\hat{\zeta})$, country i's share of BSY's cached equilibrium total $\sum_k g_k^{HW} = 0.312441$. "Efficient policy incidence" is $g_i(s^*)/v(\mathcal{N})$, the share of the surplus that accrues to i at the surplus-maximizing policy before any reallocation. The fairness gap is $\Gamma_i^{\text{raw}} = \phi_i(v)/v(\mathcal{N}) - s_i^{HW}(\hat{\zeta})$ (Definition 2). $T_i = \phi_i(v) - g_i(s^*)$ is the transfer required to move from that incidence to the benchmark, in the same units as $v(\mathcal{N})$; positive means the country receives.

Stability must be checked, not inferred. The usual route to stability is closed here. The game is not convex, and it is not even superadditive: 20 of 301 disjoint-pair tests fail, and single-player monotonicity of the cooperative game w fails in 17 of 192 additions, with a worst margin of −0.026. The failures are not high-order artifacts. They begin at the pairwise level — the United States and Canada together are worth less cooperatively than the two apart, because Canada's unilaterally profitable cut degrades United States terms of trade by more than it gains Canada — and Canada is present in 16 of the 17 violations. The worst case is the United States joining {EU, Canada}, where the coalition's value falls from 0.030 to 0.004 — and its cooperative value from +0.024 to −0.002: at that optimum the European Union captures the gains from Canada's market opening while the United States, once inside, bears the external cost of it.

We therefore verify the two properties that matter directly. *Individual rationality*: $\phi_i(v) \geq v(\{i\})$ holds at every player, with a minimum slack of 0.0186 — 2.59% of $v(\mathcal{N})$ — at Canada. *Core membership*: all 62 proper-coalition constraints are satisfied, with the same binding coalition and the same slack. The Shapley value averages marginal contributions over orderings, and that averaging absorbs a local concavity whose deepest point is 3.7% of $v(\mathcal{N})$ and which is confined to 17 of 192 additions. This is worth stating carefully because the failed sufficient condition is often used as the argument: on this economy, an argument for individual rationality or core membership that runs through monotonicity or convexity is simply wrong, and the conclusions survive only because they were checked.

Core membership is conditional in two ways that matter. First, all 62 constraints are computed with non-members held at their 1990 tariffs. That convention is not innocuous here — passive outsiders gain or lose materially from a coalition's action, and the United States' outsider gain is negative for 13 of the 31 coalitions that exclude it — and we have not shown the core result survives a γ-response convention in which outsiders best-respond (Chander 2007). Second, and more consequentially for policy, the verified core is the core of the *transferable-utility* game. The one transfer-free object the model produces — the efficient policy's own incidence, column four of Table 2 —

is not in it: Australia's incidence is negative, so Australia alone blocks the surplus-maximizing policy. Moving from that incidence to the benchmark requires a budget-balanced gross transfer of 13.5% of $v(\mathcal{N})$, with the United States, the European Union and Japan paying 2.5, 4.1 and 6.9% of the surplus respectively and Australia receiving 7.0%. The GATT operates through issue linkage and reciprocal concessions, not side payments. The correct statement is therefore conditional in both directions: the fair allocation is self-enforcing if transfers are feasible, and if they are not, even the efficient policy is blocked by a single member.

4.3. What the fairness gap measures

The gap in Table 2 is a comparison between a normative object and an estimate, and three checks are needed before it can be read as anything.

It is robust in sign and not in magnitude. Table 3 recomputes Γ under four efficient, individually rational allocation rules applied to the same $v(\cdot)$: the Shapley value, an equal split, the 1990 tariff-revenue share and the 1990 income share. The signs for the United States, Japan and Canada are the same under all four. The signs for the European Union, Australia and South Korea are not. Magnitudes are strongly rule-dependent: the United States gap ranges from +3.3 to +23.2 percentage points, a factor of seven. Any statement about the direction of the gap for the United States, Japan or Canada is a rule-independent finding; any statement about its size is a statement about the Shapley value specifically, and we never quote a magnitude without naming the rule.

Table 3: Fairness gap Γ_i (percentage points) under four efficient, individually rational allocation rules.

Rule	US	EU	Japan	Canada	Australia	S. Korea
Shapley value	+3.29	+4.30	−12.54	+2.12	+0.69	+2.14
Equal split	+6.63	−4.05	−38.81	+15.43	+14.45	+6.34
1990 revenue share	+20.98	+15.95	−39.46	+2.36	+2.41	−2.24
1990 income share	+23.21	+15.91	−32.12	+1.56	−0.79	−7.77
Sign-robust	yes	no	yes	yes	no	no

All four rules are efficient and individually rational on $v(\cdot)$; the comparator $s^{HW}(\hat{\zeta})$ is held fixed. Computed in-sample from the same coalition values as Table 2; this is a sensitivity check, not an out-of-sample test.

It measures the asymmetry the estimation found, not a distance from Shapley. Neither $\hat{\zeta}$ nor any other weight vector is privileged by the model, so we recompute the comparator at the symmetric, no-information value $\zeta = 0.5$. Four of six signs flip, and the United States gap collapses from +3.29 to −0.06 percentage points (Table 4): at a symmetric weight, the Shapley benchmark and a Horn–Wolinsky bargain award the United States almost exactly the same share. The entire United States gap at $\hat{\zeta}$ is therefore a readout of how asymmetric BSY's estimated weights are, not evidence of a structural distance between cooperative benchmarks and bargaining outcomes in general. This is a sensitivity of the comparator side only: $\phi_i(v)/v(\mathcal{N})$ does not depend on ζ at all. It is also a different question from the one Table 3 answers — that table varies the allocation rule holding $\zeta = \hat{\zeta}$ fixed — and the two checks should not be read as substitutes.

A second column of Table 4 makes the same point about the equilibrium branch rather than the weight. An independently re-solved Nash-in-Nash equilibrium at the identical $\hat{\zeta}$, converged normally, returns a materially different share vector and a correspondingly different gap; Australia and South Korea change sign, and the United States gap rises to +5.81 percentage points. The sign pattern we do report — the United States and Canada positive, Japan negative — holds on both branches, which is the reason we report signs.

Table 4: Sensitivity of the fairness gap to the bargaining weight and to the equilibrium branch.

Γ_i (pp)	US	EU	Japan	Canada	Australia	S. Korea
At $\hat{\zeta}$, cached branch (primary)	+3.29	+4.30	−12.54	+2.12	+0.69	+2.14
At $\hat{\zeta}$, alternative branch	+5.81	+10.99	−13.78	+3.04	−0.91	−5.15
At $\zeta = 0.5$, symmetric placebo	−0.06	−0.76	+6.79	+1.97	+0.47	−8.41

The benchmark side, $\phi_i(v)/v(\mathcal{N})$, is identical in all three rows; only the comparator moves. Neither branch is privileged by the model, and neither is $\hat{\zeta}$ relative to 0.5.

Most of it is a rescaling of the efficient policy's own incidence. Column four of Table 2 is the share of the surplus each country receives at the surplus-maximizing policy, before any reallocation. It correlates with the Shapley share at 0.985, and a one-parameter shrinkage of it — slope 0.752, intercept 4.13 percentage points — reproduces the benchmark to within 3.5 percentage points at every player. This cuts both ways, and we report both. In the benchmark's favor: what the Shapley axioms add on top of the physical incidence is a single describable operation, a compression of the dispersion of gains toward players with positive marginal contribution but small or negative incidence, which is precisely what symmetry and the dummy property are for — Australia and South Korea are the two players it protects. Against any strong reading of the United States gap: a referee can reproduce 97% of the cross-country pattern with that shrinkage rule alone, and the largest place where the Shapley value adds information beyond it (3.5 percentage points, at South Korea) is the same order of magnitude as the headline United States gap itself. The gap for the United States sits inside the part a simple rescaling already explains. With $n = 6$ and a two-parameter fit no significance statement is available, and we make none.

Is there a weight that would reproduce the benchmark? Definition 3 asks whether any weight vector in BSY's own box reproduces the Shapley shares. We searched from eight starting points, including $\hat{\zeta}$ and the symmetric point. The search is uninformative, and the reason is worth recording: every start exhausted its compute budget assembling its first Jacobian, because a single equilibrium evaluation takes on the order of fifteen minutes and seven perturbations are needed per step. We therefore report no answer to this question rather than a null result — the objective's differentiability was never reached — and note that the same branch-flip mechanism documented in Section 4.1 is what makes the assembly expensive in the first place.

4.4. What the bargain left on the table

BSY's estimated bargain delivers $\sum_k g_k^{HW} = 0.312$ against a grand-coalition value of $v(\mathcal{N}) = 0.720$. The level term of Eq. (14) is thus 56.6% of $v(\mathcal{N})$ and dwarfs every distributional gap in Table 2. The tempting reading — that bilateral bargaining left more than half the available surplus unrealized — is not supported without holding fixed what varies between the two numbers, and by the comparability conditions of Section 2.4 three things vary at once: coverage, instrument granularity and transferability regime.

We therefore re-solve the same optimization on BSY's own instrument. Let F^{v2} be the maximum of the members' combined welfare when each of the 151 negotiated lines can be set independently within the cuts-only box; F^{IR} the same problem with each member constrained to do at least as well as at its own 1990 threat point on that instrument; and D and D^{IR} the corresponding problems on the full 294-line schedule. All four are solved by the same MPEC machinery as Eq. (10), from multiple starts. Table 5 reports them and the decomposition they support.

Table 5: Decomposing the gap between BSY's bargained outcome and a full-schedule optimum.

	Instrument, coverage, regime	Value	Share of $v(\mathcal{N})$
$\sum_k g_k^{HW}$	BSY's bargained outcome, 151 lines	0.3124	43.4%
F^{IR}	Line-by-line, 151 lines, IR-constrained	0.4281	59.4%
F^{v2}	Line-by-line, 151 lines, unconstrained	0.4368	60.7%
D	Line-by-line, 294 lines, unconstrained	0.8850	122.9%
Decomposition of $D - \sum_k g_k^{HW} = 0.5726$			**79.51%**
Agenda coverage	$D - F^{v2}$, coverage varies only	0.4482	62.24%
Participation constraints	$F^{v2} - F^{\mathrm{IR}}$, regime varies only	0.0088	1.22%
Within-agenda shortfall	$F^{\mathrm{IR}} - \sum_k g_k^{HW}$	0.1156	16.06%

Each term varies exactly one dimension — coverage, transferability regime, or nothing — holding instrument granularity fixed at one number per tariff line. F^{IR} imposes each member's own line-by-line unilateral value as its floor; imposing a uniform zero floor instead gives 0.4285 and a within-agenda term of 16.12%, an upper bound. Components sum to 79.52% by independent rounding; the exact total is 79.51%. D exceeds $v(\mathcal{N})$ because a line-by-line instrument is strictly finer than one scalar per country: that difference, 22.9% of $v(\mathcal{N})$, is the instrument-granularity dividend at full coverage and is not part of the decomposition.

Three readings follow, and one does not.

First, the shortfall is mostly about the agenda. Extending coverage from BSY's 151 negotiated lines to the full schedule, at an unchanged line-by-line instrument, is worth 62.2% of $v(\mathcal{N})$, against 16.1% for the residual within-agenda shortfall. On the regime-matched comparison — both sides individually rational — the coverage term is 62.4% and the ratio is 3.88 to one. Widening what is negotiable is worth about four times more than bargaining harder over what already is. What the 151-line set excludes is not arbitrary: it is BSY's own estimation sample, built from reciprocal principal-supplier relationships, and agriculture leaves it by construction.

Second, transferability is not the story. The wedge between the unconstrained and individually rational line-by-line optima is 1.2% of $v(\mathcal{N})$ — the constraint removes 2.0% of the unconstrained objective. The within-agenda shortfall is essentially unchanged when every member is required to do at least as well as in 1990.

Third, the within-agenda shortfall is not an artifact of measuring participation against 1990. Replacing the 1990 threat point with BSY's own realized outcome as every member's floor — so that the comparison is against what negotiators actually delivered, not against the status quo ante — still leaves an attainable point worth 0.4109, which is 13.7% of $v(\mathcal{N})$ above the bargained outcome and preserves 85% of the within-agenda shortfall. This is a no-transfer-feasible improvement, but a *weak* one, and the qualifier is the finding: four of the six members sit exactly at their own floor, and the entire net gain accrues to Japan and Canada.

What does not follow is an attribution of the residual 16.1% to bargaining skill. Two things remain jointly embedded in it: how deep the agreed cuts were on the shared lines, and what a seven-pair bilateral architecture can deliver relative to a single grand-coalition optimizer. We attempted to separate them by re-solving BSY's own pair structure under our objective — each of the seven principal-supplier pairs maximizing its own two members' combined welfare, iterated to a fixed point. The resulting value, 0.3114, falls just below the bargained outcome it was meant to bracket (0.3124), by 0.15% of $v(\mathcal{N})$, which triggers the pre-registered boundary case rather than either conclusion. We report it as a diagnosis, not a decomposition, and note two by-products that do not depend on the failed separation: replacing a joint 151-line planner with the seven-pair architecture costs 17.4% of $v(\mathcal{N})$ on a transfer-unconstrained basis, and about a quarter (26%) of the surplus any one pair creates for itself spills, uninternalized, to the four countries outside it — a structurally analogous statement to BSY's own finding that free-rider gains account for 35.4% of the Round's realized world-welfare gain.

Two auxiliary checks support the reading. BSY's cached equilibrium moves no tariff line outside the 151-line mask — 0 of 251 off-mask cells — so "within-agenda" is a measured property of their solution and not an assumption. And their cached schedule is very nearly, but not exactly, a feasible point of our own cuts-only box: 5 of the 151 lines (3.3%) lie below zero, legal under their own lower bound of −0.25 and outside our floor of zero. The three terms above are therefore non-negative as a fact about these computed numbers, verified term by term, and not as a geometric guarantee.

4.5. The benchmark against data outside the model

Every comparison so far is model-internal: the benchmark against BSY's estimate, both computed from the same primitives. The one external check available is the tariff change that actually occurred on those lines. We compare the benchmark's own optimal line-by-line cuts to the observed 1990–2000 change in applied MFN tariffs on the identical 151 lines.

Table 6: Rank agreement with observed 1990–2000 tariff cuts, 151 negotiated lines.

Cut schedule	Spearman ρ	Lines cut
Benchmark, IR-constrained (F^{IR})	+0.641	118 of 151
Benchmark, unconstrained (F^{v2})	+0.667	—
BSY's own fitted schedule at $\hat{\zeta}$	+0.317	65 of 151
Uniform-cut placebo (model-free)	+0.714	151 of 151

Spearman rank correlation between each schedule's cut sizes and the observed 1990–2000 applied MFN change on the same lines; mid-ranks throughout. "Lines cut" counts lines moved away from the 1990 rate by more than the relevant solver tolerance; BSY's remaining 86 lines sit at their own solver's no-cut tolerance. The placebo cuts every line by the same proportion, so its rank correlation is that of the 1990 tariff level itself.

The benchmark's cuts do agree with the observed direction of liberalization ($\rho = +0.64$), and clear the band we registered in advance. They do not survive the placebo. A rule with no model in it at all — cut every line by the same proportion, which ranks lines by their 1990 tariff — scores +0.714, and a paired bootstrap on the pooled sample resolves the difference against the benchmark ($\Delta\rho = -0.073$, 95% confidence interval $[-0.148, -0.005]$). The honest statement is that the benchmark's cuts point the same way as observed liberalization but carry no information about it beyond persistence of the 1990 tariff. The same is true of BSY's fitted schedule: once rank is partialled on the 1990 tariff, neither schedule retains any measurable relationship with the observed change ($|\rho_{\mathrm{partial}}| < 0.05$ for both).

The comparison between the two models is sharper on the extensive margin than on the rank metric. BSY's cached equilibrium leaves 86 of the 151 negotiated lines — 57% — at their solver's no-cut tolerance, on lines the data show were, on net, cut over the decade; the benchmark cuts 118 of them. On the pooled sample the benchmark's rank agreement is higher, and a paired bootstrap resolves that comparison in its favor ($\Delta\rho = +0.324$, 95% CI $[+0.204, +0.459]$). But the two schedules do not have the same coverage — a schedule cannot lose rank ties on lines it never moves — and on the 65 lines where both move, the difference is not resolved ($\Delta\rho = +0.061$, 95% CI $[-0.014, +0.147]$). We take the coverage-matched comparison as the terminal one and report the head-to-head as unresolved.

Two further checks explain why the pooled and coverage-matched comparisons differ so much, and neither favors a simple story. Pinning a random 86 of the benchmark's own lines at BSY's no-cut value, 5,000 times, produces a distribution of rank correlations whose mean is +0.132 and which BSY's actual score of +0.317 exceeds at the 98.7th percentile: BSY's *choice* of which lines to leave uncut is itself informative, so the tie count alone does not explain their lower score. At the same time, the benchmark's own cut pattern resembles the model-free placebo far more closely than BSY's fitted schedule does ($\rho = 0.88$ against 0.50), so part of the benchmark's pooled advantage is its own greater resemblance to naive tariff persistence rather than superior structural content.

4.6. Registered predictions that failed

We registered the tests above before running them, and four registered claims failed. We report them here rather than only in the appendix. The United States fairness gap was registered to clear +5 percentage points; it is +3.29 on the primary branch, and clears the threshold only on the alternative branch of Table 4. The United States' passive outsider gain was registered to be negative on a majority of coalitions excluding it; it is negative on 13 of 31. The grand-coalition optimum was registered to be interior at every player; South Korea's optimum is exactly at the boundary, and only the weakened form — five of six coordinates interior — holds. Finally, an earlier version of this paper claimed that implementing the benchmark requires a transfer of about 19% of $v(\mathcal{N})$ *to* the United States, on the premise that the United States loses at the efficient policy. That claim is retracted: at the certified optimum the loser is Australia, and the United States pays into the reallocation rather than receiving from it (Table 2).

5. Robustness

5.1. A second economy

The primary panel has six similar, low-tariff economies. To check that the framework's behavior is not specific to that configuration, we repeat the exercise on the Caliendo and Parro (2015) 1993 economy, exactly enumerating coalitions from $N = 3$ (NAFTA) to $N = 7$ (NAFTA plus China, India, Brazil and Korea). This economy is heterogeneous in exactly the dimension BSY's panel is not: applied tariffs range from 3.4% to 34.5%, so unilateral gains — zero for the United States, Canada and Korea — reach \$119.7 billion for China and \$26.4 billion for India.

Two of the structural findings transfer, and two differences are informative. The Shapley allocation lies in the core here as well, verified directly at $N = 7$: all 126 proper-coalition constraints hold, with a minimum slack of 0.37% of $v(N)$ at a six-country coalition that is out of sample in the sense that it is not the grand coalition. Individual rationality holds at every N. And the failure of the sufficient conditions does *not* transfer, which is itself informative about how little such conditions travel: single-player monotonicity of w, violated 17 times in 192 additions on BSY's panel, holds on all 3,019 tested additions pooled across $N = 3$ to 7 here, and supermodularity fails on only 0.61% of triples, within a few multiples of numerical resolution. A game built from the same primitives class can be well behaved or not depending on the panel, and neither outcome can be assumed.

What does not transfer is the composition of the surplus, and this is the substantive contrast. On BSY's panel 94.05% of $v(N)$ is cooperative surplus, because almost no member gains from acting alone. On the second economy, once a high-tariff outsider enters at $N \geq 4$, only 13.1 to 16.1% is cooperative — against 75.2% at $N = 3$, where NAFTA contains no such member — and the rest is the unilateral gain high-tariff economies capture by liberalizing on their own. The raw allocation accordingly tilts toward China and India, and the cooperative allocation reverses it: the largest cooperative contributor at every $N \geq 4$ is the United States, at 35.5 to 45.2% of $w(N)$. The distinction between $\phi(v)$ and $\phi(w)$, immaterial on BSY's panel where four of six singletons are zero, is decisive here, and reading raw Shapley shares as a leverage ranking would invert the answer.

Two horse races discipline that reading further, and one of them is adverse. The raw allocation is not a relabeled GDP share (Pearson between −0.28 and +0.07 across N). It is, at $N \geq 4$, close to indistinguishable from a relabeled tariff-revenue share (Pearson 0.84 to 0.87; markedly weaker at $N = 3$, +0.29, where no high-tariff member is present). We report this because it bears directly on how much discriminating work the Shapley structure does on an economy of this shape: when one member's headroom dominates, marginal contributions track headroom. On BSY's own panel, where headroom is uniform, the same horse race is inconclusive at $n = 6$ and its band placement is driven entirely by Japan; dropping Japan moves every leg into the adverse band. Neither panel supports a claim that the Shapley allocation is doing something a simple tariff-revenue weighting could not.

5.2. Sensitivity of the headline numbers

Policy space. The cuts-only restriction $s \leq 1$ is a GATT bound-rate constraint, and it binds: four of six members would raise tariffs if allowed. Relaxing it is therefore a placebo rather than an alternative specification, and we treat it as one. The cooperative share κ falls from 94.05% under the primary convention to 10.12% when members may double their tariffs and turns negative under an unbounded space — but those figures take the numerator over an enlarged policy set while the denominator stays on $[0, 1]$, so they are not the cooperative share of any single game and are never reported as headline quantities. The lesson we do take from the placebo is the one already stated in Section 2: the benchmark values are conditional on a policy space that BSY's own solver, and the GATT, impose.

Numerical certification, and what it licenses. Multi-start optimization certifies two logically distinct things, and we separate them. For every object in Table 5 the maximized *value* agrees across starts to at least 10^{-9}. The *argmax* does not always: the full-coverage optima D and D^{IR} sit on a nearly flat ridge, with per-line differences across starts up to 0.15, while the 151-line optima F^{v2}, F^{IR} and the Pareto-dominant point agree line by line to 6×10^{-7} or better. We therefore quote D and D^{IR} only as values — they enter the decomposition and nothing else — and confine every line-level claim, including the rank comparisons of Table 6, to the objects whose argmax is pinned down. A coalition value's own solve noise is smaller still: re-solving the equilibrium at fixed recorded optima gives a relative dispersion of 7×10^{-13}, and optimizer-restart dispersion is of order 10^{-12}, so numerical error contributes nothing to any comparison in this paper at the precision we report. The uncertainty that matters is the rule dependence of Table 3 and the branch sensitivity of Table 4, both of which are orders of magnitude larger.

Equilibrium selection. Our own coalition values inherit a selection assumption symmetric to the one we document on the comparator side: $v(S)$ is defined relative to the equilibrium branch the damped fixed-point iteration selects, not to a globally unique equilibrium. Two independent monitors bound how much this matters here — agreement across damping parameters to 2.5×10^{-5} relative, and bitwise agreement of 127 of 127 independent re-solves — but the assumption is an assumption, and it is the same class of object as the branch instability that defeats the weight estimator. We report both, rather than only the comparator's.

Outsider convention. Every coalition value holds non-members at their 1990 schedule. The alternative, in which outsiders best-respond (Chander 2007), would change $v(S)$ for every proper coalition and hence the core constraints of Section 4.2. We have not run it; the results in this paper are conditional on the passive convention, and this is the largest single robustness exercise we leave open.

6. Conclusion

Quantitative models of trade negotiations divide the gains using weights estimated from the division that actually occurred. This paper asks what the same models say when the weights are removed, and answers on the economy where the question is sharpest: the panel and primitives Bagwell, Staiger, and Yurukoglu (2021) estimated their weights on. Coalitions are valued by the general-equilibrium surplus their members can generate together, and that surplus is divided by the Shapley value. Nothing in either step is estimated from the outcome being evaluated.

Three results survive the checks we can run on them. The benchmark and the estimated bargain disagree about Japan, in one direction and by a wide margin, and the sign of that disagreement — together with the positive gaps for the United States and Canada — is robust across four allocation rules and across two equilibrium branches. Its size is not: it varies by a factor of seven across rules, collapses to nothing under a symmetric bargaining weight, and is largely reproducible by a one-parameter rescaling of the efficient policy's own incidence. The Shapley allocation is in the transferable-utility core of a game that is not superadditive, verified constraint by constraint because no sufficient condition delivers it here; but the efficient policy underlying it is blocked by Australia in the absence of transfers, and reaching the benchmark requires moving 13.5% of the group's surplus between members. And the distance between what the negotiators achieved and what a full-schedule optimizer could achieve is four-fifths a question of agenda coverage — what the principal-supplier structure put on the table — and one-fifth a question of how the parties bargained over it.

Three implications follow for how such benchmarks should be used. First, raw cooperative shares are not leverage rankings. On an economy with heterogeneous protection, most of a high-tariff country's raw share is the gain it could capture unilaterally; netting that out reverses the ranking and puts the large, import-diversified economy on top. A weighting scheme built from GDP or trade volume would miss the distinction, and so would reading the raw ranking on its own terms.

Second, a fair allocation and an implementable one are different objects, and the gap between them is first-order here rather than technical. Core membership says the benchmark is self-enforcing when transfers are feasible. The GATT/WTO is by design a no-transfer institution, and on this economy the efficient policy without transfers is not merely unattractive to one member: it is blocked by one. Any normative benchmark of this type applied to a no-transfer institution owes its reader the transfer-free counterpart, and we report ours.

Third, the efficiency shortfall of an observed agreement should be decomposed before it is attributed. Comparing a full-schedule planner to a bilateral bargain crosses three boundaries at once — coverage, instrument granularity, and whether participation constraints bind — and each has to be held fixed to say anything about the others. When they are, the residual attributable to bargaining performance on the negotiated agenda is a fifth of the raw gap, and even that residual still mixes cut depth with the bilateral architecture itself, which we could not separate.

The limits of the exercise are structural, not incidental, and three of them bound what any version of this benchmark can claim. The allocation rule is not identified from tariff data, and not merely in the usual partial-identification sense: because the policy is chosen before the division, every efficient allocation rule implies exactly the same tariffs, so no tariff moment can discriminate among them. The threat point is the 1990 applied schedule, which is itself the outcome of four decades of prior bargaining, so "fairness relative to the status quo" inherits whatever the status quo encodes. And the coalition values are computed with non-members passive; a γ-response convention would change every proper-coalition value and hence the stability results, and running it is the single largest robustness exercise we leave open. To these we add two of our own making: the policy lever is one scalar per country in the benchmark itself, finer only in the line-by-line comparisons of Section 4.4, and the model has no data moment on any coalition value other than the grand coalition.

Two extensions seem worth the cost. The first is a genuine non-transferable version of the benchmark — a λ-transfer or non-transferable-utility Shapley value — which would remove the one comparability gap between the two sides that no choice of convention can close, and which the transfer result above suggests would bind. The second is the γ-response recomputation, which would tell us whether the core result is a property of this economy or of the passivity convention. Both are computationally heavy rather than conceptually hard: the machinery is the same equilibrium-constrained program, re-solved under different constraints. What the exercise already delivers is a way to ask, inside a calibrated model and without estimating a bargaining weight, what an agreement would have looked like had it divided its gains by an axiom rather than by power — and, on this panel, a specific and testable account of the ways in which it did not.

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Appendix A. Proofs

This appendix collects the formal statements and proofs of every result used in the main text, together with an explicit ledger of what is *proved*, what is *proved under a maintained hypothesis that we do not establish*, and what remains *conjectural*. The ledger is in Section A.11 at the end.

Notation reconciliation with Section 2. Notation follows the main text except for three declarations, each of which resolves a genuine collision and none of which may be dropped when this appendix is read alongside Section 2. *(i) Player set and its cardinality.* Where the main text writes N for both the player set and its cardinality, we write $\mathcal{N}$ for the set and $n \equiv |\mathcal{N}|$ for its cardinality. *(ii) The symbol n.* Section 2 uses n as a *country index* (as in $\widehat{W}_n$, τ^j_{ni}, s_n); in this appendix n denotes *only* the cardinality $|\mathcal{N}|$, and country indices are i, j, k, m, h throughout. Where an expression inherited from the main text carries a country subscript n — for instance $\widehat{\kappa}^j_{ni}$ in (7) — the reading is the main text's, and it is flagged at the point of use. *(iii) The symbol κ.* Section 2 uses $\kappa^j_{ni} = (1 + \tau^j_{ni})d^j_{ni}$ for the bilateral trade wedge and $\widehat{\kappa}^j_{ni}$ for its hat-change. In this appendix the *unsubscripted* $\kappa \equiv w(\mathcal{N})/v(\mathcal{N})$ is the *cooperative ratio* of (A.11) — a different object, appearing only from Section A.7 onward, and never carrying country–sector subscripts. To keep the two apart we also depart from the underlying derivation by renaming its corner-rate constant κ_{ij} to ρ_{ij} (Proposition A.38); κ_{ij} is not used anywhere in this appendix.

A.1. Primitives, assumptions, and the maintained hypothesis

The characteristic function (D1). For a coalition $S \subseteq \mathcal{N}$ write $\Sigma_S \equiv [0,1]^{|S|}$ for the coalition's lever box and $s_S = (s_m)_{m\in S} \in \Sigma_S$ for a lever profile; outsiders are held at the status quo, $s_m = 1$ for $m \notin S$, which we abbreviate $\mathbf{1}_{-S}$. Define the *coalitional objective*

$$\text{(A.1)} \qquad f_S(s_S) \;\equiv\; \sum_{k\in S} I_k^{\text{base}}\big[\widehat{W}_k(s_S, \mathbf{1}_{-S}) - 1\big], \qquad s_S \in \Sigma_S,$$

and the canonical (D1) characteristic function

$$\text{(A.2)} \qquad v(S) \;=\; \max_{s_S\in\Sigma_S} f_S(s_S), \qquad v(\emptyset) = 0,$$

which is eq. (8) of the main text. There is no bargaining weight, no Nash product, and no individual-rationality constraint inside (A.2). The *cooperative game* is the zero-normalization

$$\text{(A.3)} \qquad w(S) \;\equiv\; v(S) - \sum_{k\in S} v(\{k\}), \qquad w(\{k\}) = 0 \text{ by construction,}$$

and the Shapley value of a TU game u on $\mathcal{N}$ is

$$\text{(A.4)} \qquad \phi_i(u) \;=\; \sum_{S\subseteq\mathcal{N}\setminus\{i\}} \frac{|S|!\,(n-|S|-1)!}{n!}\big[u(S\cup\{i\}) - u(S)\big] \;=\; \mathbb{E}_\pi\big[u(P_\pi^{<i}\cup\{i\}) - u(P_\pi^{<i})\big],$$

the second expression being the expectation over a uniformly random arrival order π, with $P_\pi^{<i}$ the set of players preceding i.

Assumptions invoked. The assumptions are numbered as in the main text; we restate here only what the proofs actually use.

(A1) GE primitives. The world economy is the multi-country, multi-sector Eaton–Kortum model with input–output linkages of Caliendo and Parro (2015), with welfare $W_k = (w_k L_k + R_k + D_k)/P_k$ and hat-algebra system (1)–(5). The D1 objective (A.1) sums *member* welfare only; the welfare of non-players is a reported diagnostic, not part of the maximand.

(A2) Selected-equilibrium continuity, in cold-start form. For each S and each $s_S \in \Sigma_S$ the hat-algebra system admits a solution on **the branch selected by the damped fixed-point iteration cold-started from the base-year data** (with a retry schedule that lowers the damping factor from 0.2 to 0.05), and the induced map $s_S \mapsto \widehat{W}_k(s_S, \mathbf{1}_{-S})$ is continuous on Σ_S for every $k \in \mathcal{N}$ *along that selection*. This is a stipulation, not a theorem: Álvarez and Lucas (2007) is single-sector, with a tariff wedge but without intermediates or deficits, and Caliendo and Parro (2015) do not prove multi-sector uniqueness. Branch selection is in principle damping-dependent; the measured cross-damping discrepancy is 2.5×10^{-5} relative and the independent re-solve agreement is bitwise on all 127 coalitions (noise-floor calibration). Every result below that depends on A2 is stated *along the selected branch*.

(A3) Policy space. $s_m \in [0, 1]$ for members (no tariff-raising threat, no import subsidies); outsiders at $s_m = 1$ by the A4 convention. Hence Σ_S is a non-empty compact box.

(A4) Outsider convention. $v(\cdot)$ is a projection of a partition function; the projection chosen is the status quo, outsiders at $s_m = 1$.

(A5) Transferable utility within a coalition. $v(S)$ is dollar-denominated and intra-coalition transfers are feasible. Under D1 this is a *precondition for the characteristic function to have meaning*, not a convenience: if surplus were non-transferable the utilitarian sum would not represent any single allocation the coalition can reach.

(A6) Player set. The bridge's player set is the full six BSY negotiating countries $\mathcal{N}_6$; NES/ROW aggregates are non-player GE filler.

(A7) Threat point. The disagreement point is the 1990 applied tariff vector — the same convention as Bagwell, Staiger, and Yurukoglu (2021).

(A8) Share-language preconditions. (a) *raw*: $v(\mathcal{N}) > 0$ and $\phi_i(v) \geq 0$ for all i; (b) *cooperative*: $w(\mathcal{N}) > 0$ and $\phi_i(w) \geq 0$ for all i (equivalently imputation IR, $\phi_i(v) \geq v(\{i\})$); (c) *comparator side*: $\sum_k g_k > 0$ and $g_i \geq 0$ for all i. A8-b $\Rightarrow$ A8-a. If a tier fails, the corresponding object is reported in dollar-gap form via Proposition A.36 rather than as a share.

(A9) γ-equilibrium existence. In the γ-response robustness branch only, a Nash equilibrium among the outsiders exists on $[0, 1]^{|\mathcal{N} \setminus S|}$. Not proved; a stipulation confined to the robustness fork.

The one hypothesis we cannot discharge — and, on the primary panel, have refuted. Several results below — imputation IR, strict positivity of cooperative surplus for non-null players, and the upper half of the fairness-gap interval — require that adding a single player never destroys cooperative surplus. We isolate it once, name it, and carry it explicitly in the statement of every result that uses it. Its empirical status is *split across the two economies*: maintained (supported at 100%) on the second economy, **refuted by counterexample on the primary panel** — see Proposition A.2(iii′); primary-panel results that would otherwise rest on it are instead established by direct verification.

Maintained Hypothesis A.1 (Single-player w-monotonicity). *For every $S \subseteq \mathcal{N}$ and every $i \notin S$,*

$$(A.5) \qquad w(S \cup \{i\}) \ \geq \ w(S), \qquad \textit{equivalently} \qquad \Delta_i^v(S) \equiv v(S \cup \{i\}) - v(S) \ \geq \ v(\{i\}).$$

Remark A.2 (Status of Proposition A.1: empirically supported, not proved). Four points must travel with every use of Proposition A.1. *(i) It is not general superadditivity.* (A.5) is the restriction of $v(S \cup T) \geq v(S) + v(T)$ to singleton $T = \{i\}$; it is implied by, but strictly weaker than, superadditivity, and the two must not be joined by "≡". *(ii) It is not implied by any result in this paper.* In particular the replication lower bound of Lemma A12 does not deliver it: its w-analogue is generically a large negative number for high-tariff countries (Corollary A14). *(iii) It is empirically supported at* 100% *on the current D1 value tables* — 2059/2059 single-player additions at $N = 7$ and 3019/3019 pooled over $N = 3, \dots, 7$, with the companion general superadditivity check passing 966/966 at $N = 7$ (monotonicity and supermodularity sweep). The binding margins exceed the measured numerical noise floor $\varepsilon_B \approx 2.5 \times 10^{-5}$ relative – converting to dollars via $\varepsilon_B \cdot |v(\mathcal{N})|$, e.g. ≈ \$4.46M at $N = 7$ ($v(\mathcal{N}) = \$178.4$B) – by **two to three orders of magnitude**, not five: the $N = 7$ cushion of \$0.683B clears it by ≈ 153× and the global binding $N = 3$ cushion of \$0.266B clears its own ($N = 3$-scaled) floor of ≈ \$0.195M by ≈ 1363× (noise-floor calibration) — both multiples clear the 100× structural-evidence threshold declared in Proposition A.22 (the two remarks apply one standard, not two). This is strong evidence and is not a proof; Proposition A.1 is therefore a *maintained hypothesis* throughout, and every result conditioned on it says so in its own statement. *(iii′) Refutation on the primary panel; this supersedes any reading of (iii) as holding across both economies.* Point (iii) is a statement about **the second economy**. On **the primary panel** (the BSY 1990 six-country economy, cuts-only $s \in [0, 1]^6$ per the same-policy-space condition), the full 63-coalition continuous-optimization sweep *refutes* Proposition A.1: 17/192 single-player additions violate (A.5) (min margin −0.026338, i.e. 3.7% of $v(\mathcal{N}_6)$) and general superadditivity fails 20/301, with the failure already present at the pairwise level, $w(\{US, CAN\}) = -0.005591$. **Retracted claim: an earlier version of this remark asserted "every violation has US and Canada in the same coalition without Japan or South Korea."** That was the initial, qualitative signature only, and it is refuted as a universal claim by the full 192-addition census: 8 of the 17 violations fire *without* it (e.g. Canada joining {South Korea, US}). The tighter, single-country characterization that survives is that **Canada is present in** 16 **of the** 17 **violations** — only US joining {Australia, EU} has no Canada anywhere in the coalition (coalition-addition census, on the coalition values of the primary-panel solve). Consequently *no primary-panel result may be argued via Proposition A.1*; the results that would otherwise rest on it survive there by *direct verification* — $\phi_i(v) \geq v(\{i\})$ at every player with minimum slack 0.018632 (2.59% of $v(\mathcal{N}_6)$) and $w(\mathcal{N}_6) = 0.677 > 0$ — which suffices because the hypothesis is sufficient, not necessary, for those conclusions (point (i)). The theoretical status therefore sharpens from "unproven" to **"false in general for this class of games"**: the primary panel furnishes counterexamples, and the second economy's 100% support is a property of that economy (a small bloc facing passive outsiders), not of the model class.

A.2. Well-definedness of $v(\cdot)$

PROPOSITION A.3 (Existence, finiteness, sign, and the empty-coalition convention). *Assume A1–A4 and fix the equilibrium selection of A2. Then for every non-empty $S \subseteq \mathcal{N}$:*

(a) (Existence) *the program* (A.2) *attains its maximum: there exists $s_S^* \in \Sigma_S$ with $f_S(s_S^*) = v(S)$, so $v(S)$ is a maximum and not merely a supremum;*

(b) (Finiteness and sign) *there is $M_S < \infty$ with $0 \le v(S) \le \big(\sum_{k \in S} I_k^{base}\big) M_S < \infty$; the lower bound is* constructive *and* exact*;*

(c) (Convention) $v(\emptyset) = 0$*, matching the characteristic-function-form convention of Shapley (1953).*

All statements hold along the branch selected in A2.

PROOF OF PROPOSITION A.3. *Step 1 (the feasible set is compact).* By A3 the lever box is $\Sigma_S = [0,1]^{|S|}$, a non-empty closed bounded subset of $\mathbb{R}^{|S|}$, hence compact by Heine–Borel. Note that no individual-rationality constraint carves a subset out of Σ_S: under D1 the whole box is feasible, which is what removes the ϵ-buffer construction that the retired Nash-product definition required.

Step 2 (the objective is continuous). By A2, for each $k \in S$ the map $s_S \mapsto \widehat{W}_k(s_S, \mathbf{1}_{-S})$ is continuous on Σ_S along the selected branch. The objective f_S in (A.1) is a finite linear combination of these maps with constant coefficients I_k^{base}, hence continuous on Σ_S. Two features of (A.1) matter here and are worth stating because they are exactly what the retired Nash-product objective lacked: f_S contains no logarithm, so there is no singularity at a boundary of a feasible set, and f_S is additive rather than multiplicative, so it does not vanish identically on a face.

Step 3 (existence). A continuous real function on a non-empty compact set attains its maximum (Weierstrass). Applying this to f_S on Σ_S gives (a).

Step 4 (bounds). By Step 2 and compactness, each $\widehat{W}_k(\cdot)$ is bounded on Σ_S: there is $M_S < \infty$ with $|\widehat{W}_k(s_S) - 1| \le M_S$ for all $s_S \in \Sigma_S$ and $k \in S$. Hence $v(S) = f_S(s_S^*) \le \sum_{k \in S} I_k^{\text{base}} M_S < \infty$. For the lower bound, the status-quo lever $s_S = \mathbf{1}$ lies in Σ_S by A3 and reproduces the base-year tariff vector exactly; the identity counterfactual returns $\widehat{W}_k = 1$ for every k, so $f_S(\mathbf{1}) = 0$ *exactly*. Since s_S^* maximizes f_S, $v(S) = f_S(s_S^*) \ge f_S(\mathbf{1}) = 0$. This is a constructive non-negativity argument: it uses a named feasible point, not a sign restriction on the primitives. Its exactness — that $f_S(\mathbf{1})$ is 0 and not merely ≈ 0 — is enforced numerically by assertion (the identity-counterfactual check), the identity counterfactual returning $\max|\widehat{W} - 1| = 0.00\text{e}{+}00$ (noise-floor calibration). This proves (b).

Step 5 (empty coalition). For $S = \emptyset$ there are no decision variables and (A.2) is vacuous; we adopt $v(\emptyset) = 0$ as a definitional convention, which is the convention under which the Shapley formula (A.4) is stated. □

COROLLARY A4 (The value is selection-free; argmax-valued objects are not). *Under the hypotheses of Proposition A.3, the argmax correspondence $S \mapsto \arg\max_{\Sigma_S} f_S$ may be multi-valued, but every maximizer attains the same value $v(S)$; hence $v(\cdot)$, and therefore $w(\cdot)$, $\phi(v)$, $\phi(w)$ and every object built from them, are independent of which maximizer a solver returns. Objects that are functions of the maximizing* point *— the outsider welfare table $\{\widehat{W}_m(s_S^*)\}_{m \notin S}$, the non-player terms-of-trade diagnostic, and the right-hand side of Lemma A12 — inherit an ordinary numerical branch dependence, which must be reported with the solver protocol.*

PROOF OF COROLLARY A4. Immediate from the definition of arg max: if s^*, s^{**} both maximize f_S then

$f_S(s^*) = f_S(s^{**}) = v(S)$. The contrast with the retired construction is the substantive content: there the reported number was the dollar sum f_S evaluated at the maximizer of a *different* (Nash-product) objective, so two maximizers tying on the Nash product could report different dollar values. Under D1 objective and payoff coincide, and the indeterminacy disappears. □

Remark A.5 (A benefit of the cold-start form of A2). The cold-start form of A2 makes $\widehat{W}$ a function of the *world lever vector alone*: the solver is re-initialized from the base-year data at every s and never warm-started along a path. Two proofs below rely on precisely this. First, $v(S)$ is well defined as a function of S: had the branch been selected by homotopy *within* Σ_S, the selected equilibrium at a given world tariff vector could in principle have depended on the path, hence on which coalition was optimizing. Second, the replication step in Lemma A12 compares $\widehat{W}_k$ at the *same* world tariff vector reached from two different programs; equality there is an identity under A2 in cold-start form, whereas under a homotopy reading it would have required an additional path-independence argument. The cost is that branch selection is damping-dependent in principle; the live monitors are multi-damping agreement (2.5×10^{-5} relative) and independent re-solve agreement (127/127 bitwise), both recorded in the noise-floor calibration.

A.3. Bang-bang structure and exactness of the grid

The D1 program has no interior optimum anywhere in the current evidence: the persisted census records 127/127 coalitions with every coordinate at a bound (448/448 coordinates, 409 at $s = 0$ and 39 at $s = 1$, zero interior, at tolerance 10^{-6}; coordinate census). The following proposition states the structural condition under which that pattern is the *necessary* form of the solution, and derives the payoff that matters computationally: grid search over a grid containing the vertices is then exhaustive and exact.

Definition A.1 (Coordinatewise strict monotonicity). *Coalition S satisfies condition **BB** if for every member $m \in S$ and every fixed profile $s_{-m} \in [0,1]^{|S|-1}$ of the other members' levers, the section $s_m \mapsto f_S(s_m, s_{-m})$ is strictly monotone on* $[0,1]$. *The direction of monotonicity is allowed to depend on m and on s_{-m}. Condition **BB**$^+$ strengthens BB by requiring that, for each m, the* direction *of monotonicity does not depend on s_{-m}.*

Proposition A.6 (Bang-bang optima and exactness of an endpoint-containing grid). *Assume A1–A4, fix the A2 selection, and let $V_S \equiv \{0,1\}^{|S|}$ denote the vertex set of Σ_S. If S satisfies BB, then:*

(a) every maximizer of (A.2) *lies in V_S; no interior or mixed optimum exists;*

(b) $v(S) = \max_{s \in V_S} f_S(s)$; consequently, for any *finite grid $G \subseteq \Sigma_S$ with $V_S \subseteq G$, the grid-search value $\max_{s\in G} f_S(s)$ equals $v(S)$ exactly, so grid search is exhaustive and exact and no local polish step can improve on it;*

(c) if in addition S satisfies BB$^+$, the maximizer is unique and is the vertex $s^* = (s^*_m)_{m\in S}$ with $s^*_m = 0$ when f_S is decreasing in s_m and $s^*_m = 1$ when it is increasing — i.e. the optimum is determined coordinatewise, and the model's content is participation *(which members liberalize) rather than* concession depth.

Proof of Proposition A.6. *Step 1 (existence).* A maximizer exists by Proposition A.3(a); let s^* be one.

Step 2 (no interior coordinate). Suppose, for contradiction, that $s^*_m \in (0,1)$ for some $m \in S$. By BB the section $g(\cdot) \equiv f_S(\cdot, s^*_{-m})$ is strictly monotone on $[0,1]$. If g is strictly increasing then $g(1) > g(s^*_m)$; if strictly decreasing then $g(0) > g(s^*_m)$. Either way there is a feasible point — namely s^* with coordinate m replaced by the corresponding endpoint — with strictly higher objective value, contradicting optimality of s^*. Hence $s^*_m \in \{0, 1\}$ for every $m \in S$, i.e. $s^* \in V_S$, proving (a).

Step 3 (grid exactness). By (a) the maximum over Σ_S is attained on V_S, so $v(S) = \max_{\Sigma_S} f_S = \max_{V_S} f_S$. If $V_S \subseteq G \subseteq \Sigma_S$ then $\max_{V_S} f_S \leq \max_G f_S \leq \max_{\Sigma_S} f_S$, and the two ends coincide, giving $\max_G f_S = v(S)$. This proves (b). Note that (b) converts what is otherwise a discretization-error concern into a closure argument: under BB the discretization is not an approximation at all.

Step 4 (uniqueness under BB⁺). Under BB⁺ the direction of monotonicity in coordinate m is a property of m alone. Write $e_m = 1$ if f_S is increasing in s_m and $e_m = 0$ if decreasing. Step 2 applied at any maximizer forces $s^*_m = e_m$ for every m, so the maximizer is the single vertex $(e_m)_{m \in S}$, proving (c). □

Remark A.7 (Status of condition BB: swept for singletons, inferred for $|S| \geq 2$ — **second economy only; refuted on the primary panel**). We do *not* claim BB is verified for $|S| \geq 2$. What is on record is the following, and the distinction is deliberate. **Scope correction.** everything in this remark below the mechanism paragraph is evidence about *the second economy* (the seven-player Caliendo–Parro economy). On *the primary panel* (the BSY six-country 1990 economy) condition BB is **refuted**, not merely unswept, and the mechanism argument below does not survive as a general argument — see the primary-panel paragraph after the mechanism paragraph.

Mechanism. The economic reason to expect BB is that the utilitarian aggregate internalizes intra-coalition terms-of-trade transfers: member m's cut moves rents between m and its coalition partners, and those movements cancel inside $\sum_{k \in S} I^{\text{base}}_k (\widehat{W}_k - 1)$, leaving a net coalition-level effect whose sign is plausibly constant over the box. A member therefore either liberalizes fully or not at all. This mechanism is refuted as a general argument by the primary-panel evidence below; it survives only as a fact about *the second economy's own calibration*, not as a structural property of D1 objectives in general. The economic reason the two economies differ is itself instructive (Bagwell and Staiger 1999; Ossa 2014): the second economy's seven players are a small minority of the 31-region modelled world, so a member's terms-of-trade losses from full liberalization spill mostly onto non-player regions, leaving the coalition-level objective monotone in each member's own lever (bang-bang); the primary panel's six players are themselves the bulk of the modelled 1990 world economy, so intra-coalition terms-of-trade transfers largely cancel and what remains is the coalition's market power over a small non-player fringe — the classical result that a bloc with market power retains a strictly positive optimal external tariff, i.e. an interior optimum. This is a genuine two-economy contrast turning on the coalition's share of world economic size, not an inconsistency between the two economies.

Counterexamples on the primary panel (refutation, not merely non-confirmation). There the refutation is carried by the *certified continuous optimum*: the grand-coalition optimum is the strongly asymmetric interior point on five of six coordinates — **Korea sits exactly at the zero boundary, $s^*_{\text{Korea}} = 0$; the registered strong form of prediction P-19 failed on that coordinate, and "interior" here is the post-hoc weak form** — $s^* = [0.533, 0.472, 0.027, 0.326, 0.111, 0.000]$ with $v(\mathcal{N}_6) = +0.720163$, beating the best vertex of $\{0, 1\}^6$ (+0.613241) by 17.4% (primary-panel solve: 63/63 coalitions converged, quasi-Newton optimization with exact implicit-function gradients); the continuous singleton optima confirm Japan ($s^* = 0.514$) and Canada ($s^* = 0.652$) interior. The original detection — the uniform probe $s = 0.25^6$ at +0.644756 already beating every vertex, with all 144 solves converged (the vertex screen, since superseded on the values though not on the finding) — is retained as the historical first evidence; the continuous optimum strengthens it strictly. Condition BB's mechanism paragraph above is therefore refuted as a *general* argument for D1 objectives: it does not predict the primary panel, where two of six singletons are strictly interior on a resolution (Canada's $s^* \approx 0.65$) that a 5-node grid would have missed entirely. **Any future computation of $v(S)$ on the primary panel must use continuous optimization, not vertex census or grid search**; a vertex-based method would silently understate $v(S)$ there.

Direct evidence, singletons (second economy): an eleven-node recheck, not a corner-sign check. An earlier version of this remark reported only that six of seven of its singletons were "signed at $s = 0$" at a coarse resolution. That evidence is superseded by a stronger result: a dedicated 11-node (0.1-resolution) recheck of every second-economy singleton and three grand-coalition axes leaves the vertex census intact — all seven singletons endpoint-optimal (USA and Canada and Korea at $s = 1$; Mexico, China, India and Brazil at $s = 0$) and non-monotone-interior-free, and all three grand-coalition axes (USA, China, India) monotone non-increasing with $\arg\max s = 0$, all 110 equilibrium solves converged, and the reading rules pre-registered *before* the run (the eleven-node recheck). Per that recheck's own pre-registered scope note: a clean pass at 11 nodes does not *prove* BB (fine structure between nodes could still be non-monotone; an interior win at any node would have refuted it outright, and none did) — so Proposition A.6's conditionality on the current second-economy evidence still stands, but the evidentiary base is now a resolution an order of magnitude finer than the corner-sign check it replaces, at exactly the resolution (0.1) that was sufficient to expose the primary panel's Canada counterexample. The dollar values at the coarser corners agree with the finer sweep — positive at $s = 0$ for Mexico (\$1.94B), China (\$119.68B) and Brazil (\$1.67B), negative at $s = 0$ for the USA (−\$2.21B), Canada (−\$0.13B) and Korea (−\$0.38B), whence $v(\{\text{USA}\}) = v(\{\text{Canada}\}) = v(\{\text{Korea}\}) = 0$ (singleton boundary diagnostic, six-player row; these six values agree exactly with the current value tables). The seventh singleton, India, was swept over nine nodes in the same table, but that sweep predates the damping fix: three of its nine nodes failed to converge and its recorded values belong to the quarantined chain (the sweep's \$17.071B and \$5.927B against the post-fix $v(\{\text{India}\}) = \$26.411\text{B}$). It is therefore cited here as **pre-fix directional evidence only** — it indicates a monotone decreasing profile but cannot be used as a converged sweep. A damped re-sweep of the India singleton (nine equilibrium solves) remains to be run; the 11-node recheck above does not cover India (it targets the six singletons and axes that were already corner-consistent at coarse resolution), so India remains the one second-economy singleton without either coarse- or fine-resolution converged sweep evidence.

Direct evidence, $|S| \geq 2$: what the executed grand-coalition sweep does and does not establish. A one-dimensional sweep at the $N = 7$ recorded optimum has been run (grand-coalition axis sweep): two axes (USA and China), 11 nodes each on $\{0, 0.1, \dots, 1\}$, the other six members held at $s = 0$, 22 GE solves, both axes strictly monotone non-increasing with zero violations and $\arg\max s = 0$ on both (f_N falls from \$178.354B at $s_{\text{USA}} = 0$ to \$169.873B at $s_{\text{USA}} = 1$, and to \$46.784B at $s_{\text{China}} = 1$). What this establishes is precise and worth having: *the recorded vertex $s^* = \mathbf{0}$ is a global maximum along two of the seven grand-coalition axes*, so the optimizer's argmax is not a resolution artifact on those axes. What it does *not* establish is condition BB for $|S| \geq 2$. Definition A.1 quantifies over *every* member $m \in S$ and *every* fixed profile $s_{-m} \in [0, 1]^{|S|-1}$; the sweep fixes one profile ($s_{-m} = \mathbf{0}$), for two members, in one coalition out of 127 — two of 448 recorded coordinates, each at a single point of a continuum. BB for $|S| \geq 2$ therefore remains **inferred from the mechanism, not swept**, and we state it in that scope.

Indirect evidence. The full census is consistent with BB at every size: 127/127 coalitions are at vertices, including 35 of size 3, 35 of size 4, and the grand coalition, and the interior grid node $s = 0.5$ — which was on the grid and was selected 22 times before the solver fix — is selected zero times afterwards. Consistency is not implication: a vertex optimum can also arise without strict monotonicity (for instance if f_S is quasi-convex in a coordinate), so the census supports BB but does not establish it.

What would close it, and what follows meanwhile. No finite sweep can discharge BB, because the condition quantifies over a continuum of profiles s_{-m} for each of $|S|$ members: sweeps can only accumulate axis-wise counterexample-free evidence, and a proof would require signing $\partial f_S/\partial s_m$ over the whole box, which A2's bare continuity does not support (see Proposition A.8). Two things follow, and the draft must respect both. First, Proposition A.6 is used throughout as a *conditional* structural statement; the census and the two swept axes are its empirical counterparts, never its proof. Second — and this is the operative consequence — Proposition A.6(b)'s grid-exactness corollary, which is what would license reporting grid-3 values as exact rather than approximate, inherits that conditionality in full. Until BB is established for $|S| \geq 2$, the correct manuscript wording is that the grid contains every vertex and therefore returns the exact value *under BB*, with the cross-grid agreement (grid 3 versus grid 5, identical on all non-India coalitions) reported as corroborating evidence. The unconditional sentence "the discretization is not an approximation at all" is not available on the current evidence.

Remark A.8 (No interior first-order condition exists). Because all 448 recorded lever coordinates sit at bounds, the D1 program has no interior stationary point anywhere in the evidence, and there is consequently no interior first-order condition on which to hang an implicit-function-theorem comparative-statics apparatus at the lever level. We therefore do not derive $\partial s^*/\partial\theta$ for any primitive θ: the object does not exist. The comparative statics that remain well defined, and that the paper uses, are of four kinds: value-level statements obtained from the maximum theorem and the envelope property of (A.2); compositional statements comparing ϕ across player sets (cross-N); threat-point statements (the alternative-threat-point exercise); and statements in the bargaining-weight ζ over the level set of Proposition A.43. The cooperative ratio κ of Proposition A.23 is itself the sharpest comparative-static object in the paper, since by Corollary A27 it measures how much any conclusion can depend on the choice of allocation rule.

A.4. Normalization and the exact unilateral–cooperative decomposition

PROPOSITION A.9 (Singleton values and the cooperative normalization). *Assume A1–A4 and fix the A2 selection. Then:*

(a) $v(\{k\}) \geq 0$ *for every* k*, with* $v(\{k\})$ *equal to* k*'s welfare gain from its own unilateral optimal liberalization;*

(b) $v(\{k\}) = 0$ exactly *if and only if* $s_k = 1$ *(no cut) is a maximizer of the singleton program; in particular a country whose applied schedule is already at or below its unilateral optimum has* $v(\{k\}) = 0$ *exactly, not approximately;*

(c) the cooperative game w *in* (A.3) *is well defined, satisfies* $w(\emptyset) = w(\{k\}) = 0$ *for all* k*, and delivers the exact level decomposition* $v(\mathcal{N}) = \sum_{k\in N} v(\{k\}) + w(\mathcal{N})$.

PROOF OF PROPOSITION A.9. (a) is Proposition A.3(b) applied to $S = \{k\}$, together with the reading of the singleton program: with $S = \{k\}$ the only lever is k's own, all other countries are at the status quo, and the objective is $I_k^{\text{base}}(\widehat{W}_k - 1)$, i.e. k's own welfare gain.

(b) The "if" direction: if $s_k = 1$ is a maximizer, then $v(\{k\}) = f_{\{k\}}(1) = 0$ exactly by Step 4 of the proof of Proposition A.3. The "only if" direction: if $v(\{k\}) = 0$ then $s_k = 1$ attains the maximum, since $f_{\{k\}}(1) = 0 = v(\{k\})$, so it is a maximizer. The exactness in both directions is the content of the identity-counterfactual check — the identity counterfactual must return exactly zero welfare change, an invariant now enforced by assertion. On the current value tables $v(\{\text{USA}\}) = v(\{\text{Canada}\}) = v(\{\text{Korea}\}) = \$0.000\text{B}$ exactly, while $v(\{\text{Mexico}\}) = \$1.941\text{B}$, $v(\{\text{Brazil}\}) = \$1.669\text{B}$, $v(\{\text{India}\}) = \$26.411\text{B}$ and $v(\{\text{China}\}) = \$119.681\text{B}$ (monotonicity and supermodularity sweep, $N = 7$ panel).

(c) w is a finite difference of finite numbers by Proposition A.3(b), hence well defined; $w(\{k\}) = v(\{k\}) - v(\{k\}) = 0$ by construction and $w(\emptyset) = 0$; evaluating (A.3) at $S = \mathcal{N}$ and rearranging gives the level decomposition. □

PROPOSITION A.10 (Exact unilateral–cooperative decomposition of the Shapley value). *Let a denote the additive game $a(S) = \sum_{k \in S} v(\{k\})$, so that $w = v - a$. Then for every $i \in \mathcal{N}$*

$$\phi_i(v) = v(\{i\}) + \phi_i(w), \qquad \sum_{i \in \mathcal{N}} \phi_i(w) = w(\mathcal{N}), \qquad \sum_{i \in \mathcal{N}} \phi_i(v) = v(\mathcal{N}). \tag{A.6}$$

The decomposition is exact, not approximate, and holds regardless of whether v is monotone, superadditive or convex.

PROOF OF PROPOSITION A.10. *Step 1 (Shapley value of an additive game).* For the additive game a and any $S \not\ni i$, $a(S \cup \{i\}) - a(S) = \sum_{k \in S \cup \{i\}} v(\{k\}) - \sum_{k \in S} v(\{k\}) = v(\{i\})$, a constant independent of S. Substituting into (A.4),

$$\phi_i(a) = v(\{i\}) \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} = v(\{i\}),$$

because the Shapley weights are the probabilities of the events $\{P_\pi^{<i} = S\}$ under a uniform random order and therefore sum to one.

Step 2 (linearity). The Shapley value is linear in the game: from (A.4), $\phi_i(\cdot)$ is a fixed finite linear combination of the game's values, so $\phi_i(v) = \phi_i(a + w) = \phi_i(a) + \phi_i(w)$. Combining with Step 1 gives the first identity in (A.6). (Linearity here is the algebraic property of the formula; the additivity *axiom* is the normative statement that jointly with efficiency, symmetry and the dummy property characterizes ϕ uniquely (Shapley 1953).)

Step 3 (efficiency). Summing (A.4) over i and using the random-order representation, $\sum_i \phi_i(u) = u(\mathcal{N})$ for any game u with $u(\emptyset) = 0$, because for each fixed order π the marginal contributions telescope to $u(\mathcal{N}) - u(\emptyset)$. Applying this to $u = w$ and to $u = v$ gives the two remaining identities; the two are mutually consistent by Proposition A.9(c). □

Remark A.11 (Numerical check of efficiency). Shapley efficiency ($\sum_i \phi_i = v(\mathcal{N})$) is verified directly on the computed imputation: the efficiency gap at $N = 7$ is 3.05×10^{-5} dollars against $v(\mathcal{N}) = \$178.354\text{B}$ (the $N = 7$ solve), i.e. at the level of double-precision accumulation and far below the numerical noise floor.

A.5. Marginal contributions: the replication bound and the mechanical near-dummy

LEMMA A12 (R6 replication lower bound). *Assume A1–A4 and fix the A2 selection. For every coalition S and every $i \notin S$, let s_S^* be any maximizer of the S-program and define the* outsider gain

$$\mathrm{og}_i(S) \equiv I_i^{base}\big[\widehat{W}_i(s_S^*, \mathbf{1}_{-S}) - 1\big], \tag{A.7}$$

country i's passive welfare gain when S liberalizes optimally and i does nothing. Then

$$\Delta_i^v(S) \;=\; v(S \cup \{i\}) - v(S) \;\geq\; \mathrm{og}_i(S). \tag{A.8}$$

The inequality holds for every *choice of maximizer* s_S^**, and hence for the infimum of* og_i *over the argmax set; the notation* $\mathrm{og}_i(S)$ *suppresses a selection dependence that Corollary A4 flags and that Proposition A.16 handles explicitly.*

Proof of Lemma A12. *Step 1 (a feasible point for the larger program).* The profile $(s_S^*, s_i = 1) \in \Sigma_{S\cup\{i\}}$ is feasible for the $S \cup \{i\}$-program: it replicates S's optimal levers and adds i passively at the status quo. Hence

$$v(S \cup \{i\}) \;\geq\; \sum_{k\in S} I_k^{\text{base}}\big[\widehat{W}_k(s_S^*, s_i{=}1, \mathbf{1}_{-S\cup\{i\}}) - 1\big] \;+\; I_i^{\text{base}}\big[\widehat{W}_i(s_S^*, s_i{=}1, \mathbf{1}_{-S\cup\{i\}}) - 1\big].$$

Step 2 (the two world tariff vectors coincide). Under the S-program the world lever vector is $(s_S^*, \mathbf{1}_{-S})$, in which i — an outsider to S — carries $s_i = 1$. Under the point of Step 1 the world lever vector is $(s_S^*, s_i{=}1, \mathbf{1}_{-S\cup\{i\}})$. These are the *same* vector. By A2 in cold-start form, the selected equilibrium is a function of the world lever vector alone, so $\widehat{W}_k$ takes the same value in both, for every $k \in \mathcal{N}$. (See Proposition A.5: this step is an identity under A2 in cold-start form, whereas a homotopy-based selection would have required a separate path-independence argument.)

Step 3 (conclusion). By Step 2 the first sum in Step 1 equals $f_S(s_S^*) = v(S)$ and the second term equals $\mathrm{og}_i(S)$. Hence $v(S \cup \{i\}) \geq v(S) + \mathrm{og}_i(S)$, which is (A.8). □

Corollary A13 (Sufficient condition for monotonicity of v; the converse fails). *If* $\mathrm{og}_i(S) \geq 0$ *for every pair* (S, i) *with* $i \notin S$*, then* v *is monotone:* $\Delta_i^v(S) \geq 0$ *for all such pairs. The converse does* not *hold —* v *can be monotone at pairs where the outsider gain is negative, because* (A.8) *is a lower bound and not an equality. (An earlier draft stated this as "if and only if"; that was an error.)*

Proof of Corollary A13. Immediate from (A.8). For the failure of the converse it suffices to observe that (A.8) is an inequality derived from a single feasible point, so $\Delta_i^v(S)$ may exceed $\mathrm{og}_i(S)$ by any amount, in particular enough to be non-negative when $\mathrm{og}_i(S) < 0$. □

Corollary A14 (The w-analogue and why it has no force). *Subtracting* $v(\{i\})$ *from both sides of* (A.8),

$$\Delta_i^w(S) \;=\; \Delta_i^v(S) - v(\{i\}) \;\geq\; \mathrm{og}_i(S) - v(\{i\}). \tag{A.9}$$

For a high-tariff country the right-hand side is generically a large negative number: $v(\{i\})$ *reaches* \$26.4*B (India) and* \$119.7*B (China) while* $\mathrm{og}_i(S)$ *is a passive spillover an order of magnitude or more smaller. Hence Lemma A12* cannot *establish Proposition A.1, and every downstream result that needs single-player* w*-monotonicity must carry it as a maintained hypothesis rather than derive it here.*

Proof of Corollary A14. The identity $\Delta_i^w(S) = \Delta_i^v(S) - v(\{i\})$ follows from (A.3): $w(S \cup \{i\}) - w(S) = v(S \cup \{i\}) - v(S) - v(\{i\})$. Substituting (A.8) gives (A.9). The magnitude claim is a statement about the computed primitives, not a theorem (monotonicity and supermodularity sweep; the outsider gains themselves are tabulated in the coordinate census, where 369 of 441 outsider–coalition pairs have a positive gain and the R6 bound registers 0 violations out of 441). ◻

We now state the near-dummy result. The version in the theory notes bounded a country's marginal contribution by a term proportional to its base tariff alone; formalizing it revealed that the bound must also carry a *passive* free-riding term, which does not vanish with the country's own tariff. The corrected statement is a two-sided sandwich, and it is economically sharper: as a country's tariff headroom vanishes, its marginal contribution does not go to zero but collapses to the free-riding band.

Maintained Hypothesis A.15 (Lipschitz welfare response). *There exists $L < \infty$ such that for every coalition T, every $k \in \mathcal{N}$, and all $s, s' \in \Sigma_T$, $\left|\widehat{W}_k(s, \mathbf{1}_{-T}) - \widehat{W}_k(s', \mathbf{1}_{-T})\right| \le L\, d(s, s')$, where $d(s, s') \equiv \max_{m \in T} \max_{h,j} \left|\log \widehat{\kappa}^j_{mh}(s_m) - \log \widehat{\kappa}^j_{mh}(s'_m)\right|$ measures the policy perturbation in log tariff wedges. This strengthens A2's continuity to a uniform modulus; it holds automatically if $\widehat{W}$ is continuously differentiable along the selected branch on the compact box.*

proposition A.16 (Mechanical near-dummy: a sandwich, not a one-sided bound). *Assume A1–A4, fix the A2 selection, and assume Proposition A.15. For country i let $\bar{\tau}_i \equiv \max_{h,j} \tau^{j,0}_{ih}$ be its base applied-tariff scale and let $C \equiv \sum_{k \in N} I^{base}_k$. Write $M_S \equiv \arg\max_{\Sigma_S} f_S$ for the (non-empty, compact) set of maximizers of the S-program and define the free-riding band*

$$\omega_i \;\equiv\; \min_{S \subseteq \mathcal{N} \setminus \{i\}} \; \min_{s^*_S \in M_S} I^{base}_i \left[\widehat{W}_i(s^*_S, \mathbf{1}_{-S}) - 1\right], \qquad \Omega_i \;\equiv\; \max_{S \subseteq \mathcal{N} \setminus \{i\}} \; \max_{s_S \in \Sigma_S} I^{base}_i \left[\widehat{W}_i(s_S, \mathbf{1}_{-S}) - 1\right].$$

Selection qualifier *(required by Corollary A4): $\mathrm{og}_i(S)$ is a function of the maximizing* point*, not of $v(S)$, so it is selection-dependent whenever M_S is not a singleton; ω_i is therefore defined as the infimum over the whole argmax set, which makes it a valid lower bound under* every *selection. The inner minimum is attained because M_S is compact ($M_S = f_S^{-1}(\{v(S)\})$ is closed in the compact Σ_S by continuity of f_S) and $\widehat{W}_i$ is continuous on it. Ω_i needs no such qualifier: it already maximizes over all of Σ_S, not only over maximizers. Then for every $S \subseteq \mathcal{N} \setminus \{i\}$,*

$$\mathrm{og}_i(S) \;\le\; \Delta^v_i(S) \;\le\; \Omega_i \;+\; C\,L\, \log(1 + \bar{\tau}_i), \tag{A.10}$$

and therefore $\omega_i \le \phi_i(v) \le \Omega_i + C\,L\,\log(1 + \bar{\tau}_i)$. Consequently, as $\bar{\tau}_i \to 0$ the policy *channel of i's marginal contribution vanishes at rate $O(\bar{\tau}_i)$ and i's Shapley value is squeezed into the free-riding band $[\omega_i, \Omega_i]$. In the exact limit $\bar{\tau}_i = 0$: i's lever is inert ($\widehat{\kappa}^j_{ih}(s_i) \equiv 1$ for all s_i), $v(\{i\}) = 0$ exactly, and $\omega_i \le \Delta^v_i(S) \le \Omega_i$ for every S; if in addition i has no general-equilibrium linkage to any coalition (so $\widehat{W}_i \equiv 1$), then $\omega_i = \Omega_i = 0$ and i is an exact dummy with $\phi_i(v) = 0$.*

Proof of Proposition A.16. *Step 1 (lower bound).* This is Lemma A12.

Step 2 (the lever's reach is $O(\bar{\tau}_i)$). By the lever definition (7), $\widehat{\kappa}^j_{ih}(s_i) = (1 + s_i \tau^{j,0}_{ih})/(1 + \tau^{j,0}_{ih})$, which is increasing in s_i with $\widehat{\kappa}^j_{ih}(1) = 1$ and $\widehat{\kappa}^j_{ih}(0) = 1/(1 + \tau^{j,0}_{ih})$. Hence for any $s_i, s'_i \in [0, 1]$, $|\log \widehat{\kappa}^j_{ih}(s_i) - \log \widehat{\kappa}^j_{ih}(s'_i)| \le \log(1 + \tau^{j,0}_{ih}) \le \log(1 + \bar{\tau}_i)$. A country with no applied tariffs ($\bar{\tau}_i = 0$) therefore has a literally inert lever.

Step 3 (upper bound). Let $(s^\dagger_S, s^\dagger_i)$ maximize the $S \cup \{i\}$-program. Since $s^\dagger_S$ is feasible for the S-program,

$v(S) \ge f_S(s_S^\dagger)$, whence

$$\Delta_i^v(S) \le \sum_{k \in S} I_k^{\text{base}}\Big[\widehat{W}_k(s_S^\dagger, s_i^\dagger) - \widehat{W}_k(s_S^\dagger, s_i{=}1)\Big]$$
$$+ \; I_i^{\text{base}}\Big[\widehat{W}_i(s_S^\dagger, s_i^\dagger) - \widehat{W}_i(s_S^\dagger, s_i{=}1)\Big] \; + \; I_i^{\text{base}}\Big[\widehat{W}_i(s_S^\dagger, s_i{=}1) - 1\Big],$$

where we used Step 2 of the proof of Lemma A12 to identify $f_S(s_S^\dagger)$ with the terms evaluated at $s_i = 1$. The first two brackets compare two world tariff vectors differing only in i's lever, so by Proposition A.15 and Step 2 each is bounded in absolute value by $L \log(1 + \bar{\tau}_i)$; multiplying by the income weights and summing gives at most $C\,L\,\log(1 + \bar{\tau}_i)$. The third term is i's passive gain at $s_S^\dagger$, bounded above by Ω_i by definition. This yields (A.10).

Step 4 (from marginals to the Shapley value). By (A.4), $\phi_i(v)$ is a convex combination of the $\Delta_i^v(S)$, hence lies between their minimum and maximum, giving the displayed bounds on $\phi_i(v)$.

Step 5 (the exact limit). If $\bar{\tau}_i = 0$ then by Step 2 the map $s_i \mapsto \widehat{\kappa}_{ih}^j(s_i)$ is constant at 1, so the $S \cup \{i\}$-objective does not depend on s_i at all and equals $f_S(s_S) + I_i^{\text{base}}[\widehat{W}_i(s_S, \mathbf{1}_{-S}) - 1]$. Taking $S = \emptyset$ gives $v(\{i\}) = 0$ exactly (consistent with Proposition A.9(b): $s_i = 1$ is a maximizer, trivially, since every s_i is). For general S, maximizing the displayed sum gives $v(S \cup \{i\}) \le v(S) + \Omega_i$ and, by Lemma A12, $v(S \cup \{i\}) \ge v(S) + \mathrm{og}_i(S) \ge v(S) + \omega_i$. If moreover $\widehat{W}_i \equiv 1$ then both bounds are zero, so $v(S \cup \{i\}) = v(S)$ for all S and $\phi_i(v) = 0$ by the dummy property. □

Remark A.17 (What Proposition A.16 does and does not license). The proposition licenses the statement that a low-tariff country's small Shapley share is *mechanical* in the policy channel: it follows from the scalar lever $s_i \in [0, 1]$ acting on a small τ_i^0, not from any finding about strategic position. It does *not* license the stronger claim that such a country's share must be near zero: the free-riding band $[\omega_i, \Omega_i]$ is not bounded by $\bar{\tau}_i$ and is an empirical magnitude. On the current tables that band is small — Canada's raw share is 0.45–0.49% of $v(\mathcal{N})$ across $N = 4$–7 and Korea's is 1.37% at $N = 7$, both countries with $\bar{\tau} < 5\%$ (registered claim P-13, the $N = 4$ to 7 solves) — but this is measured, not derived. Two further caveats survive unchanged: the ranking implication is confined to the low end (the Shapley ordering and a GDP-weighted ordering disagree at the top), and τ^0 is itself the cumulative outcome of 1947–1990 GATT bargaining, so the near-dummy mechanism partly encodes historical bargaining outcomes rather than exogenous position.

A.6. Imputation individual rationality and strictly positive cooperative surplus

PROPOSITION A.18 (Imputation IR under Proposition A.1). *Assume A1–A4, fix the A2 selection, and assume Proposition A.1 (single-player w-monotonicity;* not *proved — see Proposition A.2). Then*

$$\phi_i(w) \;\ge\; 0 \quad \textit{for every } i \in \mathcal{N}, \qquad \textit{equivalently} \qquad \phi_i(v) \;\ge\; v(\{i\}),$$

which is imputation individual rationality, the second half of precondition A8-b. If in addition $w(\mathcal{N}) > 0$, then A8-b holds in full and therefore A8-a holds, so both cooperative and raw share language are well posed.

PROOF OF PROPOSITION A.18. *Step 1.* By the random-order representation in (A.4), $\phi_i(w) = \mathbb{E}_\pi[\, w(P_\pi^{<i} \cup \{i\}) - w(P_\pi^{<i})\,]$, an average of single-player marginal contributions of w with strictly positive weights summing to one.

Step 2. By Proposition A.1 every such marginal contribution is non-negative. An average of non-negative numbers is non-negative, so $\phi_i(w) \geq 0$.

Step 3. By Proposition A.10, $\phi_i(v) = v(\{i\}) + \phi_i(w)$, so $\phi_i(w) \geq 0$ is equivalent to $\phi_i(v) \geq v(\{i\})$. Together with $w(\mathcal{N}) > 0$ this is exactly A8-b; and A8-b implies A8-a because $\phi_i(v) = v(\{i\}) + \phi_i(w) \geq 0$ by Proposition A.9(a), while $v(\mathcal{N}) = \sum_k v(\{k\}) + w(\mathcal{N}) > 0$. □

Remark A.19 (Empirical status of A8-a/A8-b). A8-a and A8-b are checkable numerical propositions, and they are checked. On the D1 tables all five audited value tables pass imputation individual rationality, A8-a and A8-b, with minimum cooperative cushion $\min_i \phi_i(w)$ equal to +\$0.266B ($N = 3$, Canada), +\$0.659B ($N = 4$, Canada), +\$0.771B ($N = 5$, Canada), +\$0.614B ($N = 6$, Brazil) and +\$0.683B ($N = 7$, Brazil) (individual-rationality audit). The **global** binding cushion across all N is the N=3 value, +\$0.266B (Canada) — not the N=6 figure above, which is a local minimum only. These cushions exceed the measured noise floor by **two to three orders of magnitude**, not five (ε_B is a relative quantity; converting via $\varepsilon_B \cdot |v(\mathcal{N})|$ gives $\approx$ \$4.46M at $N = 7$ and $\approx$ \$0.195M at $N = 3$, so the $N = 7$ cushion clears by $\approx 153\times$ and the global $N = 3$ cushion by $\approx 1363\times$) (noise-floor calibration). This does not convert Proposition A.1 into a theorem; it means the hypothesis and its consequence are jointly consistent with every number we have computed under D1. Note also that the threshold $\phi_i \geq 0$ used in an earlier version of this check tests the wrong inequality: under D1 the singletons are large and positive, so $\phi_i \geq 0$ can pass while true IR fails by tens of billions. The correct threshold is $\phi_i(v) \geq v(\{i\})$ and it is what the audit above reports.

PROPOSITION A.20 (Non-null players receive strictly positive cooperative surplus). *Assume Proposition A.1 (not proved). If player i is non-null in the cooperative game — that is, there exists $S \subseteq \mathcal{N} \setminus \{i\}$ with $w(S \cup \{i\}) > w(S)$ — then*

$$\phi_i(w) \;=\; \sum_{S \subseteq \mathcal{N}\setminus\{i\}} \frac{|S|!\,(n-|S|-1)!}{n!}\big[w(S\cup\{i\}) - w(S)\big] \;>\; 0.$$

PROOF OF PROPOSITION A.20. Every Shapley weight $|S|!(n-|S|-1)!/n!$ is strictly positive. Under Proposition A.1 every bracket is non-negative, and by the non-null assumption at least one bracket is strictly positive. A weighted sum with strictly positive weights of non-negative terms, at least one of which is strictly positive, is strictly positive. □

COROLLARY A21 (Zero surplus for a non-null player is impossible under the Shapley axioms but permitted by the generalized-Nash family). *Assume Proposition A.1. Then no non-null player receives zero cooperative surplus under the Shapley imputation. By contrast, in the Horn–Wolinsky/generalized-Nash bilateral family the corner weight $\zeta \to 0^+$ drives the weak party's surplus to exactly zero at rate $O(\zeta)$ (Proposition A.38), and corner weights lie inside the admissible parameter space. The correct statement of the contrast is therefore one about* permissiveness of a solution-concept family, *not about a parameter value: the generalized-Nash family admits, at admissible parameters, an allocation that the Shapley axioms exclude for* every *game.*

PROOF OF COROLLARY A21. The first sentence is Proposition A.20. The second is Proposition A.38. The third is the conjunction: the two solution concepts differ not merely in the value of a free parameter but in whether a zero-surplus allocation to a non-null player is attainable at all. □

Remark A.22 (Convexity: what the sweep does and does not show — **second economy only**). **Scope.** This remark concerns the second economy (the seven-player Caliendo–Parro economy) throughout; supermodularity is not swept on the primary panel, whose general superadditivity outright fails $20/301$ times, a fact this remark's "convex to within numerical resolution" conclusion describes a different economy and must not be read to contradict (§2.3). Convexity (supermodularity) is a different property from Proposition A.1 and is not needed for anything above. On the D1 tables the supermodularity failure rate is 0.6% (31 of 5103 valid triples at $N = 7$; 40 of 7101 pooled; monotonicity and supermodularity sweep). **Evidence-strength threshold** (stated once here and cross-cited by Proposition A.2(iii), so the two remarks apply one standard, not two). We judge a violation-magnitude comparison against its noise floor as follows: a binding gap or margin below 20× the relevant noise floor is read as a numerical artifact, not structural evidence; above 100× it is read as structural evidence; strictly in between, we report it as inconclusive rather than asserting either reading. The correct reading under this threshold, and the only one the draft may use, is: *the game is convex to within numerical resolution; the residual* 0.61% *of violating triples have gaps of* \$0.016–\$0.046*B against a cross-configuration sensitivity of about* \$0.003–\$0.006*B, a ratio of roughly* 3×–15× *— below the* 20× *artifact threshold just stated — and is not evidence of genuine non-convexity.* Reporting 0.61% as a measured non-convexity finding would be a "significant inside the error band" claim in mirror image. Substantively this is a pre-registered FAIL of the non-convexity half of prediction P-12 (threshold: failure rate ≥ 10%), and the pre-written branch was taken and reported as a failure rather than reframed.

A.7. The fairness gap is locked into an interval of width κ

This subsection contains the paper's sharpest formal result about the positive–normative bridge. It answers a question left open in earlier rounds — whether imputation IR pins down the *sign* of the fairness gap — with a negative answer and a more useful positive one: IR does not sign Γ_i, but it confines Γ_i to an interval whose width is exactly the cooperative share of the surplus, uniformly across players and across allocation rules.

Notation. Fix the bridge player set $\mathcal{N}_6$ (A6) and assume A8-a, $v(\mathcal{N}) > 0$. Write

$$u_i \;\equiv\; \frac{v(\{i\})}{v(\mathcal{N})} \quad \text{(player } i\text{'s \textit{unilateral share})}, \qquad \kappa \;\equiv\; \frac{w(\mathcal{N})}{v(\mathcal{N})} \quad \text{(the \textit{cooperative ratio})}, \tag{A.11}$$

so that $\sum_i u_i = 1 - \kappa$ by Proposition A.9(c). Let $s_i^{HW}(\zeta, b)$ denote the comparator share of Definition 2. For an arbitrary allocation rule ψ applied to the cooperative game, write $\Gamma_i^{\text{raw}}(\psi) \equiv \big[v(\{i\}) + \psi_i(w)\big]/v(\mathcal{N}) - s_i^{HW}$, which specializes to (12) when $\psi = \phi$ by Proposition A.10.

PROPOSITION A.23 (IR interval for the fairness gap). *Assume A8-a ($v(\mathcal{N}) > 0$) and $w(\mathcal{N}) \geq 0$ (efficiency (i) forces $\sum_j \psi_j(w) = w(\mathcal{N})$ while IR (ii) forces every summand ≥ 0; if $w(\mathcal{N}) < 0$ no allocation can satisfy both, so the class of allocations the proposition quantifies over is empty and the interval is vacuous rather than proved — the hypothesis is therefore necessary, not merely convenient, and must be checked before invoking this proposition on either economy. On the second economy $w(\mathcal{N}) > 0$ at every $N \in \{3, \ldots, 7\}$ and on the primary panel $w(\mathcal{N}_6) = 0.677 > 0$ (primary-panel solve); both are verified, not assumed, when the proposition is actually invoked in this paper). Let ψ be* any *allocation of the cooperative game satisfying*

(i) efficiency*: $\sum_{j \in \mathcal{N}} \psi_j(w) = w(\mathcal{N})$;*

(ii) imputation IR*: $\psi_j(w) \geq 0$ for every $j \in \mathcal{N}$.*

Then for every i,

$$u_i - s_i^{HW} \quad\le\quad \Gamma_i^{\mathrm{raw}}(\psi) \quad\le\quad u_i + \kappa - s_i^{HW}, \tag{A.12}$$

an interval of width exactly κ — independent of i, independent of ψ, and independent of the comparator s^{HW}.

Proof of Proposition A.23. *Step 1 (lower bound on ψ_i).* By (ii) applied to $j = i$, $\psi_i(w) \ge 0$.

Step 2 (upper bound on ψ_i). By (i), $\psi_i(w) = w(\mathcal{N}) - \sum_{j\ne i}\psi_j(w)$, and by (ii) applied to each $j \ne i$ the subtracted sum is non-negative, so $\psi_i(w) \le w(\mathcal{N})$.

Step 3 (rescale). Steps 1–2 give $\psi_i(w) \in [0, w(\mathcal{N})]$. By Proposition A.10's decomposition applied to ψ (which here is a definition, $\psi_i(v) \equiv v(\{i\}) + \psi_i(w)$, and is the Shapley identity when $\psi = \phi$), dividing by $v(\mathcal{N}) > 0$ gives $\psi_i(v)/v(\mathcal{N}) \in [u_i,\, u_i + \kappa]$.

Step 4 (subtract the comparator). Subtracting s_i^{HW} from each side of Step 3's interval gives (A.12). The width is $(u_i + \kappa) - u_i = \kappa$, which involves neither i nor ψ nor s^{HW}. □

Proposition A.24 (Both endpoints are attained: the interval cannot be narrowed). *Let $n \ge 2$ and $w(\mathcal{N}) > 0$. Then within the class of allocations satisfying (i)–(ii) of Proposition A.23, both bounds in (A.12) are attained, so the interval is the exact range of $\Gamma_i^{\mathrm{raw}}(\psi)$ over that class and no strengthening is possible without an axiom beyond efficiency and IR.*

Proof of Proposition A.24. *Step 1 (lower endpoint).* Fix $j_0 \ne i$ (possible since $n \ge 2$) and let ψ^ℓ assign $\psi^\ell_{j_0}(w) = w(\mathcal{N})$ and $\psi^\ell_j(w) = 0$ for all $j \ne j_0$. Then $\sum_j \psi^\ell_j = w(\mathcal{N})$, satisfying (i), and every component is non-negative since $w(\mathcal{N}) > 0$, satisfying (ii). Its i-component is 0, so $\Gamma_i^{\mathrm{raw}}(\psi^\ell) = u_i - s_i^{HW}$, the lower bound.

Step 2 (upper endpoint). Let ψ^u assign $\psi^u_i(w) = w(\mathcal{N})$ and $\psi^u_j(w) = 0$ for $j \ne i$. The same two checks give (i)–(ii), and $\Gamma_i^{\mathrm{raw}}(\psi^u) = u_i + \kappa - s_i^{HW}$, the upper bound.

Step 3 (range). By Proposition A.23 the map $\psi \mapsto \Gamma_i^{\mathrm{raw}}(\psi)$ takes values in $[u_i - s_i^{HW}, u_i + \kappa - s_i^{HW}]$; the class (i)–(ii) is convex (it is the scaled simplex $\{\psi \ge 0 : \sum_j \psi_j = w(\mathcal{N})\}$) and Γ_i^{raw} is affine in ψ, so its image is the closed interval between the two attained endpoints, i.e. the whole interval. The Shapley value is one interior point of that set; it is not at either endpoint except in degenerate games. □

Proposition A.25 (Consistency with $\sum_i \Gamma_i = 0$). *Under efficiency of ψ and of the comparator shares ($\sum_i s_i^{HW} = 1$), $\sum_i \Gamma_i^{\mathrm{raw}}(\psi) = 0$, and this is compatible with (A.12): summing the lower bounds gives $-\kappa \le 0$ and summing the upper bounds gives $(n-1)\kappa \ge 0$.*

Proof of Proposition A.25. $\sum_i \Gamma_i^{\mathrm{raw}}(\psi) = \frac{1}{v(\mathcal{N})}\left[\sum_i v(\{i\}) + \sum_i \psi_i(w)\right] - 1 = \frac{1}{v(\mathcal{N})}\left[\sum_i v(\{i\}) + w(\mathcal{N})\right] - 1 = \frac{v(\mathcal{N})}{v(\mathcal{N})} - 1 = 0$, using Proposition A.9(c). For the bounds, $\sum_i (u_i - s_i^{HW}) = (1-\kappa) - 1 = -\kappa$ and $\sum_i (u_i + \kappa - s_i^{HW}) = (1-\kappa) + n\kappa - 1 = (n-1)\kappa$, and $-\kappa \le 0 \le (n-1)\kappa$ whenever $\kappa \ge 0$. □

Corollary A26 (Shapley-free sign conditions). *Under the hypotheses of Proposition A.23: $u_i > s_i^{HW} \Rightarrow \Gamma_i^{\mathrm{raw}} > 0$, and $u_i + \kappa < s_i^{HW} \Rightarrow \Gamma_i^{\mathrm{raw}} < 0$. Neither condition requires computing any Shapley value: they use only the singleton value $v(\{i\})$, the grand-coalition value $v(\mathcal{N})$, and the comparator share.*

Proof of Corollary A26. Immediate from the two ends of (A.12). □

Corollary A27 (Small κ makes the conclusion independent of the allocation rule). *As $\kappa \to 0$ the interval*

([A.12](#)) *collapses to the single point* $u_i - s_i^{HW}$*:* every *efficient, IR allocation rule then delivers the same fairness gap. Consequently* κ *is a direct, reportable measure of how sensitive the paper's distributional conclusion is to the choice of solution concept, and should be reported alongside* Γ *as a robustness statistic rather than treated as a descriptive aside.*

Proof of Corollary A27. The interval has width κ by Proposition A.23 and contains $u_i - s_i^{HW}$ as its left endpoint; as $\kappa \downarrow 0$ the right endpoint converges to the left. By Proposition A.24 the interval is exactly the range over the class, so the collapse is not merely of an outer bound but of the range itself. □

corollary A28 (Computation-free screen for the pre-registered Γ_{US} claim). *Suppose* $v(\{US\}) = 0$*, so* $u_{US} = 0$*. Then* $\Gamma_{US}^{\mathrm{raw}} \in [-s_{US}^{HW}, \kappa - s_{US}^{HW}]$ *for every efficient, IR allocation. Hence:*

(a) $\Gamma_{US}^{\mathrm{raw}} > 0$ *under* some *such allocation requires* $\kappa > s_{US}^{HW}$*;*

(b) $\Gamma_{US}^{\mathrm{raw}} \geq +5pp$ *requires* $\kappa \geq s_{US}^{HW} + 0.05$*.*

With the measured comparator value $s_{US}^{HW}(\hat{\zeta}) = 10.04\%$ *on the six-country denominator (pre-registered sanity checks), these read* $\kappa > 10.04\%$ *and* $\kappa \geq 15.04\%$ *respectively.*

Proof of Corollary A28. Set $u_{US} = 0$ in (A.12) and read off the two endpoints; (a) and (b) are the statements that the relevant threshold lies below the upper endpoint. Necessity is exactly Proposition A.23; the endpoint is attainable by Proposition A.24, so the conditions are necessary and, at the level of the whole allocation class, also sufficient for attainability. □

Remark A.29 (Scope of Proposition A.23 and its corollaries). Four qualifications travel with this block and none may be dropped. *(1)* Both bounds use hypothesis (ii), but asymmetrically: the lower bound uses i's own IR and the upper bound uses the IR of the *others*. If IR fails for a set of players, the upper bound relaxes to $\psi_i(w) \leq w(\mathcal{N}) - \sum_{j \neq i} \min\{\psi_j(w), 0\}$ and the lower bound to $\psi_i(w) \geq \min\{\psi_i(w), 0\}$; the width is then κ plus the total IR violation, divided by $v(\mathcal{N})$. Hypothesis (ii) for the Shapley value is Proposition A.18, whose own hypothesis Proposition A.1 is not proved; but Proposition A.23 itself is stated for an arbitrary efficient, IR allocation and therefore does not depend on the Shapley value at all. *(2)* κ and u_i must be measured on the same calibration as each other; the paper's two-economy protocol forbids importing a κ computed on the Caliendo–Parro 1993 economy into a statement about the BSY 1990 economy. The second economy's values are $\kappa = 75.2\%$ ($N = 3$), 13.4% ($N = 4$), 13.1% ($N = 5$), 13.6% ($N = 6$) and 16.1% ($N = 7$) (individual-rationality audit); these are a *prior* for the primary panel, not a substitute computation, and Corollary A28's screen must be run with the primary panel's own eight solves (seven singletons plus the grand coalition). *(3)* The result is about the raw gap Γ^{raw}. The analogous bound for the cooperative gap Γ^{coop} requires re-deriving with the aligned comparator of eq. (13) and has not been done. *(4)* Both sides of Γ^{raw} must use the same threat point (A7, the same-threat-point condition), the same denominator (the same-normalization condition), and the same player set (the same-player-set condition); in particular the six-country denominator is primary and the five-country in-pair renormalization is a robustness variant whose systematic inflation must be disclosed. *Two further comparability conditions are completed here: (4a) same policy space* (the same-policy-space condition) — the comparator $s_i^{HW}(\hat{\zeta})$ is the outcome of BSY's cuts-only bargaining problem ($s \in [0, 1]^n$), so u_i, κ, and any ψ invoked in Corollary A28 must also be computed on the cuts-only policy space; mixing a cuts-only comparator against an extended-space ($\bar{s} > 1$) $v(\mathcal{N})$ or $v(\{i\})$ is forbidden as a primary object and is reported only as the separate policy-space placebo of the policy-space placebo. *(4b) same transferability regime, or an explicit decomposition of the difference* (the transferability-regime condition) — ψ is a TU allocation of w, while s^{HW} is the outcome of BSY's NTU bilateral bargain (their own estimation holds every transfer variable at zero); the TU feasible set contains the NTU one, so Γ_i^{raw}'s sign readings inherit a known-direction bias (TU is weakly more favorable to any given i than NTU) that is not corrected by this proposition and must be disclosed alongside any Γ reported from it.

Remark A.30 (A8 checks do not automatically transfer across policy-space placebo rows). The policy-space placebo table (the policy-space placebo, $\bar{s} \in \{1, 1.5, 2, \infty\}$) recomputes $v(S)$ and $w(S)$ over a *different* feasible set at every row, so A8-a ($v(\mathcal{N}) > 0$), A8-b ($w(\mathcal{N}) \geq 0$, the hypothesis added to Proposition A.23 above), and A8-c (comparator-side positivity, $\sum_k g_k > 0$ and $g_i \geq 0$) are properties of the row's own game and are *not* inherited from the $\bar{s} \leq 1$ primary row by default. Two consequences, one cautionary and one exculpatory: *(i)* the deep-sweep row's originally reported $\kappa_A^{\text{deep}} = -2.17\%$ would, taken at face value, have meant $w(\mathcal{N}) < 0$ at that row — an A8-b violation that voids any invocation of Proposition A.23 there, not merely a cosmetic sign flip; **the corrected value satisfies $\kappa_A^{\text{deep}} \leq +8.53\%$ as a point bound (an earlier version of this statement wrote $\geq$, reversing the direction established elsewhere: the numerator $\sum_i v(\{i\})^{\text{deep}}$ is a 10-node grid *lower* bound on the true singleton values, so the ratio $\kappa = 1 - \sum_i v(\{i\})/v(\mathcal{N})$ it enters is an *upper* bound, not a lower one), and $\gtrsim +8.4\%$ under the robustness reconstruction (Japan/Canada continuous re-values plus a quadratic fit at the US/EU peaks) — which is the *lower* bound A8-b actually needs to be restored at that row**; the -2.17% was a stale-denominator artifact, not a genuine subadditivity finding at that policy space, and the row was re-checked against its own $w(\mathcal{N})$, never because A8-b was assumed to survive from $\bar{s} \leq 1$. Every future placebo row must state its own A8-a/b/c check explicitly; a row that has not been re-checked must be reported as A8-status UNVERIFIED, not silently treated as PASS. *(ii)* A8-c is the one exception to the no-inheritance rule: the comparator $s_i^{HW}(\hat{\zeta})$ is computed once, on BSY's own (cuts-only) bargaining problem, which the $v(S)$-side placebo sweep never touches — so A8-c's PASS at the primary row (pre-registered sanity checks) is unaffected by which $\bar{s}$ row is being read on the $v(S)$ side, and does not need re-verification per row.

A.8. Two players: the collapse propositions and the threat-point equivalence

Throughout this subsection $n = 2$ with players $\{i, j\}$, utility is transferable within the pair (A5), and the feasible set is the TU frontier $\{(x_i, x_j) : x_i + x_j \leq V\}$ with $V \equiv v(\{i, j\})$. The non-negativity of the pair's cooperative surplus, $w(\{i, j\}) = V - v(\{i\}) - v(\{j\}) \geq 0$ — the two-player instance of Proposition A.1, and therefore *not* established — is needed by exactly one result below and is carried in that result's own statement (Proposition A.33), not assumed silently here. Proposition A.31 and Proposition A.32 do not require it; Proposition A.34 inherits it through Proposition A.33 and says so.

PROPOSITION A.31 (Shapley on the cooperative game splits evenly). *$\phi_i(w) = \phi_j(w) = \frac{1}{2}w(\{i, j\})$, and hence $\phi_i(v) = v(\{i\}) + \frac{1}{2}w(\{i, j\})$.*

PROOF OF PROPOSITION A.31. With $n = 2$ the Shapley formula (A.4) gives $\phi_i(w) = \frac{1}{2}[w(\{i\}) - w(\emptyset)] + \frac{1}{2}[w(\{i, j\}) - w(\{j\})]$. By Proposition A.9(c), $w(\emptyset) = w(\{i\}) = w(\{j\}) = 0$, so $\phi_i(w) = \frac{1}{2}w(\{i, j\})$, symmetrically for j. Adding $v(\{i\})$ via Proposition A.10 gives the raw statement. Note that this is a property of the *cooperative* game: the corresponding statement for the raw game, $\phi_i(v) = \frac{1}{2}V$, is false whenever

$v(\{i\}) \neq v(\{j\})$ — see Proposition A.32. □

PROPOSITION A.32 (B-1a: raw side, BSY's threat point). *Let the disagreement point be the 1990 status quo, $d = (0, 0)$, matching A7 and Bagwell, Staiger, and Yurukoglu (2021). The symmetric Nash bargaining solution on the TU frontier is $x_i = x_j = \frac{1}{2}V$, and this coincides with the raw-game Shapley value $\phi_i(v)$* ***if and only if*** *$v(\{i\}) = v(\{j\})$.*

PROOF OF PROPOSITION A.32. *Step 1.* The symmetric Nash solution maximizes $x_i x_j$ subject to $x_i + x_j \leq V$, $x \geq 0$. The constraint binds at any solution (increasing either coordinate increases the product), so substituting $x_j = V - x_i$ and maximizing $x_i(V - x_i)$ gives the unique interior maximizer $x_i = V/2$, hence $x_i = x_j = V/2$.

Step 2. By Proposition A.31, $\phi_i(v) = v(\{i\}) + \frac{1}{2}\left[V - v(\{i\}) - v(\{j\})\right] = \frac{1}{2}V + \frac{1}{2}\left[v(\{i\}) - v(\{j\})\right]$.

Step 3. Therefore $\phi_i(v) = \frac{1}{2}V$ if and only if $v(\{i\}) = v(\{j\})$. The condition is not innocuous in this application: the computed singletons range from exactly \$0 (USA, Canada, Korea) to \$119.7B (China), so the coincidence generically fails. □

PROPOSITION A.33 (B-1b: cooperative side, unilateral-optimum threat point). *Let the disagreement point be each country's own unilateral optimum, $d_i = v(\{i\})$, $d_j = v(\{j\})$, and assume the pair's cooperative surplus is non-negative, $w(\{i, j\}) \geq 0$ (the two-player instance of Proposition A.1, which is* not *established — see Proposition A.2; without it the bargaining set $\{x \geq d : x_i + x_j \leq V\}$ is empty and the solution is undefined). Then the symmetric Nash bargaining solution on the TU frontier is $x_i = v(\{i\}) + \frac{1}{2}w(\{i, j\}) = \phi_i(v)$, with no symmetry condition on the singletons required — "unconditional" here means unconditional* in the singletons*, in contrast with Proposition A.32's iff, not free of the non-emptiness hypothesis just stated.*

PROOF OF PROPOSITION A.33. The symmetric Nash solution maximizes $(x_i - d_i)(x_j - d_j)$ subject to $x_i + x_j \leq V$ and $x \geq d$. Writing $y_k \equiv x_k - d_k \geq 0$, the constraint becomes $y_i + y_j \leq V - d_i - d_j = w(\{i, j\})$, a non-negative budget by the stated hypothesis. By the argument of Step 1 of Proposition A.32 applied in y-coordinates, the solution is $y_i = y_j = \frac{1}{2}w(\{i, j\})$, i.e. $x_i = v(\{i\}) + \frac{1}{2}w(\{i, j\})$, which equals $\phi_i(v)$ by Proposition A.31. □

PROPOSITION A.34 ($R8$/B-2: zero-normalization is a threat-point redefinition). *Comparing Proposition A.32 and Proposition A.33 (and inheriting the latter's hypothesis $w(\{i, j\}) \geq 0$): at $n = 2$ with transferable utility and a linear frontier, computing the Shapley value on the zero-normalized game w instead of on v is* exactly *equivalent to moving the disagreement point from "1990 tariffs frozen" to "every country at its own unilateral optimum". For general n the corresponding statement is the identity of Proposition A.10, $\phi_i(v) = v(\{i\}) + \phi_i(w)$, read as the axiomatic expression of a two-stage procedure: each country first takes its unilateral dividend, then the residual cooperative surplus is divided by the Shapley axioms.*

PROOF OF PROPOSITION A.34. *Step 1 ($n = 2$).* Proposition A.33 shows that the symmetric Nash solution at $d = (v(\{i\}), v(\{j\}))$ equals $\phi_i(v) = v(\{i\}) + \phi_i(w)$, which is the w-Shapley allocation translated back to v-space. Proposition A.32 shows that the symmetric Nash solution at $d = (0, 0)$ equals the raw-Shapley allocation only under the singleton-symmetry condition. Hence the operation "zero-normalize, then apply Shapley" is the same operation as "move the disagreement point to the unilateral optima, then apply the symmetric Nash bargaining solution". *Step 2 (general n).* Proposition A.10 is an identity, so the two-stage reading is available for any n; but the equivalence with a *bargaining* solution is established here only for $n = 2$ with a linear frontier, and we do not claim it beyond that. □

COROLLARY A35 (Why the raw game is the bridge's primary operand). *Three independent reasons, each established above.* (1) Threat-point comparability (A7). *The comparator gains $g_i(\zeta, b)$ are measured from the 1990 status quo with nothing subtracted; $\phi_i(w)/w(\mathcal{N})$ has already netted out $v(\{i\})$. By Proposition A.34 the cooperative object implicitly uses a different disagreement point, so only $\phi_i(v)/v(\mathcal{N})$ and s_i^{HW} share a threat point (the same-threat-point condition).* (2) Behavioral credibility. *The disagreement point of Proposition A.33 is a counterfactual — if China could unilaterally capture* \$119.7*B, the question of why it did not is unanswered by the model — whereas the 1990 applied schedule is an observed historical fact.* (3) Weaker preconditions. *Raw share language needs only A8-a, and A8-b $\Rightarrow$ A8-a strictly (Proposition A.18), so the raw bridge survives a failure of the maintained hypothesis that would invalidate the cooperative one. The cooperative gap is not thereby deprecated: it answers a well-posed and independently interesting question, and is reported as a secondary object with its differing threat point stated.*

PROOF OF COROLLARY A35. (1) is Proposition A.34 plus the definition of g_i; (3) is the last sentence of Proposition A.18; (2) is an interpretive statement about the primitives, recorded here because it is load-bearing for the choice of primary object and must not be relegated to a footnote. □

A.9. The dollar identity and the inverse map

PROPOSITION A.36 ($D4'$.0: exact dollar identity). *Fix ζ and an equilibrium branch b, let $g_i(\zeta, b)$ be the comparator's dollar welfare gains on the player set $\mathcal{N}_6$, and suppose $\sum_k g_k(\zeta, b) \neq 0$ so that $s_i^{HW} = g_i / \sum_k g_k$ is defined. Then for every i,*

$$\text{(A.13)} \qquad \underbrace{\phi_i(v) - g_i(\zeta, b)}_{\textit{dollar gap } D_i} = \underbrace{v(\mathcal{N})\,\Gamma_i^{\text{raw}}(\zeta, b)}_{\textit{distributional term}} + \underbrace{s_i^{HW}(\zeta, b)\Big[v(\mathcal{N}) - \textstyle\sum_k g_k(\zeta, b)\Big]}_{\textit{share-weighted efficiency term}},$$

and, summing over i, $\sum_i D_i = v(\mathcal{N}) - \sum_k g_k(\zeta, b)$.

PROOF OF PROPOSITION A.36. *Step 1.* Expand the right-hand side using $\Gamma_i^{\text{raw}} = \phi_i(v)/v(\mathcal{N}) - s_i^{HW}$:

$$v(\mathcal{N})\left[\frac{\phi_i(v)}{v(\mathcal{N})} - s_i^{HW}\right] + s_i^{HW} v(\mathcal{N}) - s_i^{HW} \sum_k g_k = \phi_i(v) - s_i^{HW} v(\mathcal{N}) + s_i^{HW} v(\mathcal{N}) - s_i^{HW} \sum_k g_k.$$

Step 2. The two $s_i^{HW} v(\mathcal{N})$ terms cancel, leaving $\phi_i(v) - s_i^{HW} \sum_k g_k$. By the definition of s_i^{HW}, $s_i^{HW} \sum_k g_k = g_i$, so the right-hand side equals $\phi_i(v) - g_i = D_i$. *Step 3.* Summing (A.13) over i and using $\sum_i \Gamma_i^{\text{raw}} = 0$ (Proposition A.25) and $\sum_i s_i^{HW} = 1$ gives $\sum_i D_i = 0 + \big[v(\mathcal{N}) - \sum_k g_k\big]$. □

Remark A.37 (Two uses of Proposition A.36). First, it welds the distributional claim to the efficiency comparison: a reported Γ^{raw} finding is a distributional statement precisely because the level difference $v(\mathcal{N}) - \sum_k g_k$ is carried in a separate, explicitly displayed term, so a share comparison cannot silently smuggle in a level comparison. Second, it is the fallback metric when share language is not available: (A.13) is defined for gaps of either sign, so if A8-a or A8-b fails the corresponding object is reported in dollars. In the degenerate limit $\sum_k g_k \to 0$ the shares themselves become numerically unstable and the wording must say so rather than divide by a near-zero denominator.

PROPOSITION A.38 ($D3.1$: corner weights force zero surplus, at rate $O(\zeta)$). *Consider the bilateral generalized-Nash program* $\max\{\zeta \log g_i + (1-\zeta)\log g_j\}$ *over the Pareto frontier* $\mathcal{F}_{ij}$*, subject to* $g_i, g_j \geq 0$*, and assume the frontier regularity condition A-D3.1:* $\mathcal{F}_{ij}$ *is non-empty, compact, has a point with both coordinates strictly positive, and its Pareto boundary is described by a strictly decreasing, strictly concave, continuously differentiable* $g_j = g_j^P(g_i)$*, with* $g_j^{P\prime}(0) \neq 0$ *(a non-degenerate frontier slope at the corner). Then the solution* $(g_i^*(\zeta), g_j^*(\zeta))$ *is unique and satisfies*

$$\lim_{\zeta\to 0^+} g_i^*(\zeta) = 0, \qquad g_i^*(\zeta) = \zeta\,\rho_{ij} + o(\zeta), \qquad \rho_{ij} \;=\; \frac{g_j^P(0)}{\left|g_j^{P\prime}(0)\right|} \;\in\; (0,\infty).$$

The clause $g_j^{P\prime}(0) \neq 0$ *is what makes* ρ_{ij} *finite: it is implied by strict monotonicity of* g_j^P *only in the weak sense* $g_j^{P\prime}(0) \leq 0$*, so it must be assumed separately. If it fails (a frontier that is flat at the corner), the convergence* $g_i^*(\zeta) \to 0$ *of the first display still holds, but the rate is slower than* $O(\zeta)$ *and the constant is not defined. The rate constant is written* ρ_{ij}*, not* κ_{ij} *as in the theory note, to avoid collision with the cooperative ratio* κ *of* (A.11) *and with the main text's trade wedge* κ_{ni}^j*; see the notation reconciliation in Section A.*

PROOF OF PROPOSITION A.38. *Step 1 (uniqueness and first-order condition).* On the Pareto boundary the objective is $\Lambda(g_i) \equiv \zeta \log g_i + (1-\zeta)\log g_j^P(g_i)$. Since log is increasing and concave and g_j^P is concave, $\log g_j^P$ is concave; $\log g_i$ is concave; so Λ is strictly concave on the interior of its domain and the maximizer is unique. Its first-order condition is

$$\frac{\zeta}{g_i} \;=\; (1-\zeta)\,\frac{\left|g_j^{P\prime}(g_i)\right|}{g_j^P(g_i)}. \tag{A.14}$$

Step 2 (the limit). As $\zeta \to 0^+$ the left-hand side of (A.14) tends to 0 for any fixed $g_i > 0$, while the right-hand side tends to the strictly positive number $|g_j^{P\prime}(g_i)|/g_j^P(g_i)$. The equality can therefore be maintained only if $g_i \to 0$, so that the left-hand side stays bounded away from 0. Equivalently, the iso-objective curves $g_i^\zeta g_j^{1-\zeta}$ flatten toward horizontal lines as $\zeta \to 0$, and the tangency point slides to the corner $\arg\max\{g_j : g_i \geq 0\}$, at which $g_i = 0$.

Step 3 (rate). Rearranging (A.14),

$$g_i^*(\zeta) = \frac{\zeta}{1-\zeta}\cdot\frac{g_j^P\big(g_i^*(\zeta)\big)}{\left|g_j^{P\prime}\big(g_i^*(\zeta)\big)\right|}.$$

By Step 2, $g_i^*(\zeta) \to 0$; by continuity of g_j^P and $g_j^{P\prime}$ at 0 *together with* $g_j^{P\prime}(0) \neq 0$ (which keeps the denominator bounded away from zero in a neighbourhood of the corner), the ratio on the right converges to the finite positive limit $g_j^P(0)/|g_j^{P\prime}(0)| = \rho_{ij}$, and $\zeta/(1-\zeta) = \zeta + o(\zeta)$. Hence $g_i^*(\zeta) = \zeta\rho_{ij} + o(\zeta)$. □

Remark A.39 ($D3.1a'$: corners break standard inference, not just arithmetic). Three of the comparator's seven estimated weights sit exactly on the optimizer bounds $\{0.005, 0.995\}$, with standard errors as large as 0.13–0.47. The right description of the consequence is not "numerical ill-conditioning" but a genuine inferential discontinuity: when a parameter lies on the boundary of its admissible space, the limiting distribution of the M-estimator is not the usual normal and delta-method confidence intervals are invalid (Andrews 1999). Two things follow.

First, individual-rationality verdicts near a corner are hypersensitive to small errors in the disagreement point — this is the numerical symptom. Second, and more fundamentally, confidence sets for ζ near a corner must be built by a method valid on the boundary, which is why the inverse map in Definition 3 is reported as a Chernozhukov–Hong–Tamer-style (Chernozhukov, Hong, and Tamer 2007) level set rather than as a delta-method ellipsoid. The two choices are responses to the same phenomenon, not two independent conventions.

PROPOSITION A.40 (D4.1: symmetric two-player inverse map). *Let $n = 2$ with a single bargaining pair and suppose the Pareto frontier is symmetric, $(g_i, g_j) \in \mathcal{F}_{ij} \iff (g_j, g_i) \in \mathcal{F}_{ij}$, and that the generalized-Nash solution is unique for each ζ (as in Proposition A.38). Then $\zeta^* = \frac{1}{2}$ solves the inverse-map equation exactly, with zero residual: $s_i^{HW}(\frac{1}{2}) = \frac{1}{2} = \phi_i(w)/w(\mathcal{N})$.*

PROOF OF PROPOSITION A.40. *Step 1 (comparator side).* Relabelling $i \leftrightarrow j$ maps the program with weight ζ on i into the program with weight $1 - \zeta$ on i, and leaves $\mathcal{F}_{ij}$ invariant by symmetry. Uniqueness of the solution then forces $g_i^*(1-\zeta) = g_j^*(\zeta)$ for every ζ. Setting $\zeta = \frac{1}{2}$ gives $g_i^*(\frac{1}{2}) = g_j^*(\frac{1}{2})$ and hence $s_i^{HW}(\frac{1}{2}) = g_i^*/(g_i^* + g_j^*) = \frac{1}{2}$.

Step 2 (normative side). By Proposition A.31, $\phi_i(w)/w(\mathcal{N}) = \frac{1}{2}$.

Step 3. The two sides coincide at $\zeta = \frac{1}{2}$. At $n = 2$ the raw and cooperative criteria agree here because Propositions A.31 and A.32 both give a symmetric $\frac{1}{2}$ split when $v(\{i\}) = v(\{j\})$ (which holds trivially at $n = 2$ with identical players by the symmetry hypothesis), so $L(\zeta) = \|s^{HW}(\zeta) - \phi(v)/v(\mathcal{N})\|_2^2$ — the raw-side criterion of Definition 3, used uniformly throughout this paper (an earlier draft of this step used the cooperative-side criterion $\|s^{HW}(\zeta) - \phi(w)/w(\mathcal{N})\|^2$ instead; at general n these are different objects with different normative content, per §"Why the raw game is primary" in the main text, and only the raw-side definition is used from Proposition A.43 onward, so this step is restated with the same definition for consistency) — attains its global minimum value 0 there. This is a consistency check on the machinery, not new content: the general inverse map, applied to the symmetric two-player case, recovers exactly the split that Proposition A.31 gives directly. ◻

PROPOSITION A.41 (D4.2: asymmetric two-player inverse map by frontier-ratio matching). *Let $n = 2$ and let the Pareto frontier be $g_j = G(g_i)$ on $[0, \bar{g}]$ with G continuously differentiable, $G' < 0$, $G'' < 0$, $G(0) > 0$ and $G(\bar{g}) = 0$. Let $\rho \equiv \phi_i(v)/\phi_j(v) > 0$ be the target ratio (raw tier; the cooperative tier replaces $\phi(v)$ by $\phi(w)$ throughout). Then:*

(a) there is a unique *$\hat{g}_i \in (0, \bar{g})$ with $G(\hat{g}_i)/\hat{g}_i = 1/\rho$;*

(b) the weight that makes $\hat{g}_i$ the generalized-Nash solution is

$$\zeta^* = \frac{|G'(\hat{g}_i)|\,\hat{g}_i}{|G'(\hat{g}_i)|\,\hat{g}_i + G(\hat{g}_i)}; \tag{A.15}$$

(c) in the transferable-utility limit $G(g_i) = \bar{G} - g_i$ (so $|G'| \equiv 1$), (A.15) collapses to $\zeta^* = \hat{g}_i/\bar{G} = \phi_i(v)/[\phi_i(v) + \phi_j(v)]$, which equals the raw Shapley share provided *the frontier total is the raw total, $\bar{G} = v(\mathcal{N})$; with the cooperative total $\bar{G} = w(\mathcal{N})$ it equals the cooperative share instead. The two tiers must be declared separately and never mixed.*

PROOF OF PROPOSITION A.41. *Step 1 (existence and uniqueness of the ratio-matching point).* Let $h(g) \equiv G(g)/g$ on $(0, \bar{g}]$. Then h is continuous, and $h(g) \to +\infty$ as $g \to 0^+$ (since $G(0) > 0$), while $h(\bar{g}) = 0$. By the intermediate value theorem h attains the value $1/\rho > 0$. Moreover $h'(g) = [G'(g)g - G(g)]/g^2 < 0$ on $(0, \bar{g})$

because $G' < 0$ and $G > 0$ there; hence h is strictly decreasing and the solution $\hat{g}_i$ is unique. (Only $G' < 0$ and $G > 0$ are needed for this step; strict concavity is used in Step 2, not here.)

Step 2 (the tangency condition at an on-frontier point). For fixed ζ, the generalized-Nash program on the frontier maximizes $\Lambda(g) = \zeta \log g + (1-\zeta) \log G(g)$. As in Step 1 of Proposition A.38, $G'' < 0$ makes Λ strictly concave, so the first-order condition $\zeta/g = (1-\zeta)|G'(g)|/G(g)$ is necessary and sufficient. Solving for ζ at $g = \hat{g}_i$, $\zeta/(1-\zeta) = |G'(\hat{g}_i)|\hat{g}_i/G(\hat{g}_i)$, which rearranges to (A.15). The essential point is *where* the condition is evaluated: at the on-frontier point $\hat{g}_i$ constructed in Step 1, not at the Shapley dollar pair (ϕ_i, ϕ_j), which in general does not lie on $\mathcal{F}_{ij}$ — substituting an off-frontier point into a condition derived on the frontier is invalid, and an earlier draft did exactly that.

Step 3 (TU limit). With $G(g) = \bar{G} - g$ the ratio-matching equation $(\bar{G} - \hat{g}_i)/\hat{g}_i = \phi_j/\phi_i$ gives $\hat{g}_i = \bar{G}\,\phi_i/(\phi_i + \phi_j)$; substituting $|G'| = 1$ and $G(\hat{g}_i) = \bar{G} - \hat{g}_i$ into (A.15) gives $\zeta^* = \hat{g}_i/\bar{G} = \phi_i/(\phi_i + \phi_j)$. With $\bar{G} = v(\mathcal{N})$ and $n = 2$, $\phi_i + \phi_j = v(\mathcal{N})$ by efficiency, so $\zeta^* = \phi_i(v)/v(\mathcal{N})$. □

Remark A.42 (The frontier slope is a baseline correction, not a robustness check). Proposition A.41(c) shows that ζ^* equals a Shapley share only in the TU limit $|G'| = 1$. The comparator's own bargaining problems are non-transferable-utility, and the single available measurement of the frontier slope — the Japan–Korea frontier of the benchmark paper, $|G'| \approx 1.96$ — is 96% away from the TU limit. This is adverse evidence for assumption A5, it is reported as such, and it is why the $|G'|$ wedge correction enters every ζ^* and Γ conversion by default, with the TU limit shown only as a reference column. The slope $G'(\hat{g}_i)$ has no closed form and must be estimated numerically by perturbing ζ near the candidate solution and re-solving; that computation is not performed here.

PROPOSITION A.43 (*D*4.3: existence of a best-fit weight vector). *Assume A-D4.1: for ζ in the compact box $\mathcal{Z} = [\underline{\zeta}, \overline{\zeta}]^K$ the comparator's best-response solver converges to a gains vector $g(\zeta)$ that is continuous in ζ along the selected branch, and $\sum_k g_k(\zeta) > 0$ on $\mathcal{Z}$. Then $L(\zeta) = \left\| s^{HW}(\zeta) - \phi(v)/v(\mathcal{N}) \right\|_2^2$ is continuous on $\mathcal{Z}$ and attains its minimum: a best-fit $\zeta^* \in \arg\min_{\mathcal{Z}} L$ exists. The claim is unconditional in the sense that it does not require* $\min L = 0$.

Proof of Proposition A.43. By A-D4.1 each g_k is continuous on $\mathcal{Z}$ and $\sum_k g_k > 0$ there, so $s^{HW} = g/\sum_k g_k$ is continuous (quotient of continuous functions with non-vanishing denominator) and maps into the simplex. The target $\phi(v)/v(\mathcal{N})$ is a fixed vector, so L is a composition of continuous maps, hence continuous. $\mathcal{Z}$ is a non-empty compact box. By Weierstrass, L attains its minimum. (This is the same argument as Proposition A.3(a); every "continuous objective on a compact box" claim in this paper runs through it.) □

Remark A.44 (Exact solvability and the dimension count). Exact solvability, $L(\zeta^*) = 0$, is strictly stronger than Proposition A.43 and requires the Shapley target to lie in the image $s^{HW}(\mathcal{Z})$. Without further knowledge of that image no general argument delivers it, and we treat it as a case-by-case numerical question, diagnosed *ex post* by the residual. On the dimension count, with K estimated pairs and target simplex dimension $n-1$: if $K < n-1$ there is generically no exact solution and only a best fit; if $K = n-1$ an exact solution, when it exists, is generically isolated — locally unique precisely when the Jacobian $Ds^{HW}(\zeta^*)$ has full rank, a checkable local condition, not a global uniqueness theorem; if $K > n-1$ — the actual configuration, $K = 7$ against $n - 1 = 5$ — the inverse problem is underdetermined and the exact-solution set is generically a manifold of dimension $K - (n-1) = 2$, so a selection rule (the Mahalanobis regularization of Definition 3) is required for a point-valued answer. The earlier proposal to restrict the player set to the five in-pair countries in order to define away an unreachable coordinate is *not* available: the country absent from every pair has a non-zero general-equilibrium welfare gain, so its coordinate is reachable, and the source of underdetermination is the dimension count alone.

Conjecture A.45 (Uniqueness of the inverse map at general n). *For the actual configuration ($K = 7 > n-1 = 5$) the level set Z_0^* of exact solutions, when non-empty, contains a unique point at which the Mahalanobis criterion of Definition 3 is minimized, and the multi-start numerical solver converges to it from every admissible starting value. We do not prove this, and we do not have evidence for it: L is not known to be convex, s^{HW} is not known to be monotone in ζ, and multiple local minima are not excluded. The claim is to be resolved numerically — by reporting the residual $L(\zeta^*)$, the Jacobian condition number at the solution, and the dispersion across at least five starting values — and must be labelled a conjecture wherever it appears in the text until then.*

A.10. The outsider convention

The baseline value $v_{sq}(S)$ holds outsiders at the status quo (A4). The γ-response alternative $v_\gamma(S)$ replaces the outsider block by a Nash equilibrium played *among the outsiders*,

$$s_m^\gamma(s_S) \in \arg\max_{s_m} \widehat{W}_m\big(s_S, s_m, s^\gamma_{-S-m}(s_S)\big) \qquad \forall m \notin S \text{ simultaneously,}$$

$$v_\gamma(S) = \max_{s_S \in \Sigma_S} \sum_{k \in S} I_k^{\text{base}}\big[\widehat{W}_k\big(s_S, s^\gamma_{-S}(s_S)\big) - 1\big],$$

whose existence is assumption A9 and is not proved. Under D1 both conventions use the *same* utilitarian objective, so any difference between them is attributable to the convention alone.

proposition A.46 (Equality cases). *Assume A1–A3, A9, and fix the A2 selection. (a)* (Joint-maximizer hypothesis, stated explicitly, not derived.) *Suppose there exists $s_S^* \in \Sigma_S$ that simultaneously maximizes* both *programs — the status-quo program (objective $\sum_{k\in S} I_k^{base}[\widehat{W}_k(s_S, \mathbf{1}_{-S}) - 1]$) and the γ-response program (objective $\sum_{k\in S} I_k^{base}[\widehat{W}_k(s_S, s^\gamma_{-S}(s_S)) - 1]$) — and that at this point the outsider block coincides with the status quo, $\mathbf{1}_{-S} = s^\gamma_{-S}(s_S^*)$. This is an assumption about the joint optimizer, not a consequence of the outsider-coincidence condition alone (the two programs can still differ at every other point of Σ_S). Under this hypothesis, $v_\gamma(S) = v_{sq}(S)$. (b)* (Dummy outsiders, corrected sufficient condition.) *If every outsider $m \notin S$ has **zero total general-equilibrium dependence** on s_S — $\widehat{W}_m(s_S, \cdot) \equiv \widehat{W}_m$ identically in s_S, holding the rest of the system's response fixed — then m's best response does not depend on s_S, so $s^\gamma_{-S}(s_S)$ is constant in s_S; if in addition the status quo is that constant, $v_\gamma(S) = v_{sq}(S)$ identically. **Zero bilateral trade shares ($\pi^j_{mk} = \pi^j_{km} = 0$ for all $k \in S, j$) are*** **not** ***sufficient for this**: in a fully-coupled multi-sector CP input–output economy, m's welfare can still respond to S's liberalization through third-country price and wage general-equilibrium channels even absent any direct bilateral trade link. The in-house counterexample is Canada: it trades bilaterally with none of the six BSY bargaining pairs, yet its measured general-equilibrium welfare gain is strictly positive and non-negligible ($g_{Canada} = +0.00387$ native units at $\hat{\zeta}$, $\approx 1.24\%$ of the six-country total, pre-registered sanity checks). Zero bilateral share is therefore* not *a usable proxy for the zero-total-derivative hypothesis this proposition actually needs.*

Proof of Proposition A.46. (a) Under the joint-maximizer hypothesis, $v_\gamma(S)$ equals the γ-program's objective evaluated at s_S^*, and $v_{sq}(S)$ equals the status-quo program's objective evaluated at the same s_S^*; the two objectives coincide pointwise at s_S^* because $\mathbf{1}_{-S} = s_{-S}^{\gamma}(s_S^*)$ makes the outsider block identical in both, so the two values are equal. This is a direct consequence of the stated hypothesis, not an independent derivation — the substantive content of part (a) is entirely in identifying when the joint-maximizer hypothesis is plausible (e.g. when the γ-program's objective is globally identical to the status-quo program's, which is the special case where outsiders never react at all, subsuming part (b)). (b) Under the corrected zero-total-derivative hypothesis, m's maximization problem does not involve s_S and its solution is a constant in s_S. This is the dummy-outsider case, and it links back to the dummy property of the imputation (the dummy property). The corrected hypothesis is strictly stronger than, and not implied by, the zero-bilateral-share condition an earlier draft used. □

Remark A.47 (No a priori ordering: honest status). It is tempting to state "neither $v_{sq}(S) \geq v_\gamma(S)$ nor the reverse holds uniformly" as a proposition. We do not, and the reason is specific. Two mechanisms point in opposite directions: (i) if the coalition's liberalization makes further own liberalization privately optimal for an outsider, the outsider cuts, coalition members gain market access, and $v_\gamma > v_{sq}$; (ii) if the coalition's liberalization erodes an outsider's terms of trade, the outsider raises its tariff to recapture rents, and $v_\gamma < v_{sq}$. But what actually signs the comparison is the *level gap* $\tau_m^\gamma(\tau_S^*) - \tau_m^{sq}$, not the derivative $\partial\tau_m^\gamma/\partial\tau_S$; and under the standard premise that applied tariffs sit below each country's own optimal-tariff schedule, that level gap has a predictable sign, which is in tension with a clean "no ranking" claim. We have not resolved that tension, and constructing a two-sided possibility result would require explicit examples in both directions, which we do not have. This is therefore recorded as a *remark* motivating a numerical γ-response robustness run — the two mechanisms are candidate directions to be adjudicated coalition by coalition — and it carries no proposition-level weight anywhere in the paper. The only proved statements about the comparison are the equality cases of Proposition A.46.

A.11. Ledger: what is proved, what is conditional, what is conjectural

The four open items, stated plainly. *First*, Proposition A.1 (single-player w-monotonicity) is not proved and cannot be obtained from Lemma A12; it is supported at 100% (3019/3019 pooled) **on the second economy**, with margins about two to three orders of magnitude above the measured noise floor (an earlier version of this statement said "five orders of magnitude," comparing a relative quantity to dollar cushions without converting units; see Proposition A.2), **and is refuted on the primary panel** (17/192; Proposition A.2(iii′) — primary-panel results survive by direct verification, not by this hypothesis), and everything that depends on it — imputation IR, strictly positive cooperative surplus, the $n = 2$ cooperative collapse, and the upper half of the fairness-gap interval *when the allocation is the Shapley value* — is stated conditionally. (The interval itself, Proposition A.23, is stated for the whole efficient-plus-IR class and does *not* inherit it; but a claim about Γ_i under the *Shapley* allocation, such as the P-07′/P-07″ screen of Corollary A28, does, since it needs $\phi_j(w) \geq 0$ for $j \neq i$.) *Second*, condition BB underlying Proposition A.6 is signed at $s = 0$ for six of seven singletons and inferred from the mechanism for $|S| \geq 2$; the 127/127 vertex census and the two swept grand-coalition axes are its empirical counterparts, and no finite sweep can discharge it (Proposition A.7). *Third*, Proposition A.15 (uniform Lipschitz welfare response) is declared, not established: A2 supplies continuity only, and Proposition A.16 is conditional on the strengthening. *Fourth*, uniqueness of the inverse map at general n is a conjecture with no supporting argument (Proposition A.45). Nothing in the paper's headline claims requires resolving the fourth; the first three are stated as hypotheses in the propositions that use them, and the numerical evidence for each is cited where it is used.

Table A.1: Status of every formal result in this appendix: the value function, the imputation, and the marginal-contribution results. "Conditional" means proved, given a hypothesis that this paper does not establish; the hypothesis is named in the result's own statement, never in a footnote.

Result	Content	Status	Leans on
Proposition A.3	$v(S)$ exists, finite, ≥ 0; $v(\emptyset) = 0$	Proved	A1–A4; A2 branch
Corollary A4	value is selection-free	Proved	Proposition A.3
Proposition A.6	BB $\Rightarrow$ vertex optima; grid exact	**Conditional** (on BB)	condition BB, unswept for $\lvert S\rvert \geq 2$
Proposition A.9	singletons ≥ 0; $= 0$ exactly at $s^* = 1$	Proved	A1–A4; the identity-counterfactual check
Proposition A.10	$\phi_i(v) = v(\{i\}) + \phi_i(w)$	Proved (exact)	Shapley linearity
Lemma A12	replication lower bound	Proved	A2 in cold-start form
Corollary A13	sufficient condition for v-monotonicity	Proved (converse false)	Lemma A12
Corollary A14	w-analogue has no force	Proved	Lemma A12
Proposition A.16	near-dummy sandwich	**Conditional**	Proposition A.15 (declared, not established)
Proposition A.18	$\phi_i(w) \geq 0$, i.e. imputation individual rationality/A8-b	**Conditional**	Proposition A.1 (second economy); *on the primary panel the hypothesis is refuted — conclusion holds by direct verification, Proposition A.2(iii′)*
Proposition A.20	non-null $\Rightarrow \phi_i(w) > 0$	**Conditional**	Proposition A.1 (second economy); primary panel: premise refuted, conclusion by direct verification
Corollary A21	permissiveness contrast	**Conditional**	Proposition A.1 (second economy); primary panel: premise refuted, conclusion by direct verification

Table A.2: Status of every formal result in this appendix, continued: the fairness-gap interval, the two-player collapse, and the inverse map.

Result	Content	Status	Leans on
Proposition A.23	Γ_i interval of width κ	Proved	A8-a; *and* $w(\mathcal{N}) \geq 0$ (necessary, not merely convenient: the efficiency+IR class is empty otherwise); efficiency+IR class
Proposition A.24	both endpoints attained	Proved	$n \geq 2$, $w(\mathcal{N}) > 0$
Proposition A.25	$\sum_i \Gamma_i = 0$ compatible	Proved	efficiency
Corollary A26, Corollary A27, Corollary A28	sign screen; κ-collapse; κ threshold	Proved	Proposition A.23
Proposition A.31	n=2 even split on w	Proved	—
Proposition A.32	B-1a, iff $v(\{i\}) = v(\{j\})$	Proved	TU, linear frontier
Proposition A.33	B-1b, unconditional *in the singletons*	**Conditional**	TU, linear frontier, $w(\{i, j\}) \geq 0$ (= Proposition A.1 at n=2)
Proposition A.34	$R8$: normalization ≡ threat point	**Conditional**, at n=2	TU, linear frontier; inherits $w(\{i, j\}) \geq 0$
Proposition A.36	$D4'.0$ dollar identity	Proved (algebra)	$\sum_k g_k \neq 0$
Proposition A.38	corner ζ: zero surplus, rate ρ_{ij}	Proved	A-D3.1 + $g_j^{P\prime}(0) \neq 0$
Proposition A.40	$\zeta^* = 1/2$ symmetric n=2	Proved	frontier symmetry
Proposition A.41	frontier-ratio construction	Proved	$G' < 0$, $G'' < 0$
Proposition A.43	best-fit ζ^* exists	Proved	A-D4.1
Proposition A.45	uniqueness of ζ^* at general n	**Conjecture**	—
Proposition A.46	outsider-convention equality cases	**(a) Conditional** (joint-maximizer hypothesis, stated not derived); **(b) Proved**, premise corrected to zero total GE derivative	A9 (robustness fork); (b) additionally needs the corrected zero-total-derivative condition, strictly stronger than zero bilateral trade share
Proposition A.47	no a priori ordering v_{sq} vs v_γ	**Not proved** (remark)	—

Appendix B. The Second Economy: Full Results

This appendix reports the detail behind the second-economy robustness exercise summarized in Section 5.1. The economy is the 31-region, 40-sector 1993 calibration of Caliendo and Parro (2015). Coalitions are enumerated exactly from $N = 3$ (the NAFTA members) to $N = 7$ (NAFTA plus China, India, Brazil and South Korea), under the same characteristic function Eq. (8), the same cuts-only policy space, and the same passive-outsider convention as the primary panel.

B.1. Rest-of-world aggregation

The dataset reports 30 named countries plus a residual rest-of-world composite (ROW). We document its construction, and our treatment of it, because the composite is held fixed rather than admitted as a player.

Trade, value added, and gross output. For each variable X (bilateral exports, sectoral value added, or sectoral gross output), $X^{\text{ROW}} = X^{\text{world}} - \sum_{n=1}^{30} X^n$, with world totals from UN COMTRADE (trade) and UNIDO INDSTAT (production).

Input–output structure. For each pair of sectors (j, k) the coefficient γ_{ROW}^{jk} is the median of the corresponding sample-30 coefficients. The cross-country input–output structure is highly correlated in this sample (mean pairwise correlation 0.99 across the 30 countries), so the median imputation introduces negligible heterogeneity.

Tariff structure. ROW's applied tariff vector is the 8.57% aggregate average reported in the Caliendo and Parro (2015) appendix, applied uniformly across sectors. ROW's tariffs are never choice variables in any coalition.

Treatment as outsider. ROW is fixed at its data values in every computation reported here and is not a player. Admitting it would inflate $v(S \cup \{\text{ROW}\})$ mechanically, by roughly 16% of world value added, without representing any strategic decision — the same reason the "not elsewhere specified" composites are excluded from the primary panel's player set (Section 3.1).

B.2. Coalition values and the two allocations

Table B.1 reports the grand-coalition value, its cooperative component, and the extremes of the two allocations at each N. Table B.2 gives the full per-country breakdown at $N = 7$.

B.3. Is the imputation a relabeled GDP share?

A natural concern is that the imputation merely re-labels economic size. We test this at each N against the income share $I_i / \sum_{k \in S_N} I_k$ from the same calibration. If the Shapley value were a size proxy, the cross-country correlation should approach unity and the rankings should agree.

The Pearson correlation is uniformly small in absolute value ($|\rho| \leq 0.28$, largest at $N = 4$), and ranks disagree beyond the top country at every $N \geq 4$. The sharpest divergence is at $N = 4$: China holds a GDP share of 7.3% and a raw Shapley share of 89.6%, while the United States holds 81.0% of GDP and 6.1% of the raw allocation.

Table B.1: Coalition values and allocation extremes, second economy, $N = 3$ to 7.

N	$v(N)$ ($B)	$w(N)$ ($B)	$w(N)/v(N)$ (%)	Largest share: raw $\phi(v)$	Largest share: coop. $\phi(w)$	Min. IR slack (%)
3	7.81	5.87	75.2	Mexico 60.1%	US 48.6%	3.40
4	140.46	18.84	13.4	China 89.6%	US 45.2%	0.47
5	170.35	22.31	13.1	China 74.1%	US 40.1%	0.45
6	173.20	23.50	13.6	China 72.9%	US 39.2%	0.35
7	178.35	28.65	16.1	China 71.5%	US 35.5%	0.38

Values in base-year US dollars. $w(N) = v(N) - \sum_n v(\{n\})$ is the cooperative game of Eq. (9). "Min. IR slack" is $\min_n\{\phi_n(v) - v(\{n\})\}$ as a percentage of $v(N)$; it binds at Canada for $N \leq 5$ and at Brazil for $N = 6, 7$, and is positive at every N, so individual rationality holds throughout. Source: the exact coalition enumerations and the individual-rationality check of the replication package.

Table B.2: Raw and cooperative allocations at $N = 7$, second economy.

Country	Unilateral $v(\{n\})$ ($B)	Raw allocation $\phi_n(v)$ ($B)	Raw allocation share (%)	Cooperative allocation $\phi_n(w)$ ($B)	Cooperative allocation share (%)
China	119.68	127.51	71.49	7.83	27.31
India	26.41	28.49	15.98	2.08	7.27
United States	0.00	10.16	5.70	10.16	35.46
Mexico	1.94	6.53	3.66	4.59	16.01
South Korea	0.00	2.44	1.37	2.44	8.51
Brazil	1.67	2.35	1.32	0.68	2.38
Canada	0.00	0.88	0.49	0.88	3.06
Total	149.70	178.35	100.00	28.65	100.00

Rows ordered by raw share. The two orderings differ in their first three places: the raw allocation is led by the two high-tariff members, whose unilateral values alone are 82% of $v(N)$, while the cooperative allocation — which nets those values out — is led by the United States, whose unilateral value is exactly zero. This is the distinction that is immaterial on the primary panel, where four of six singletons are zero, and decisive here.

Table B.3: Raw Shapley share against GDP share, second economy.

N	Pearson	Spearman
3	+0.075	−0.500
4	−0.276	+0.600
5	−0.246	+0.100
6	−0.176	+0.086
7	−0.123	+0.071

GDP is measured by base-year household income. Source: the benchmark comparison of the replication package.

Interpretation, in both directions. Raw marginal contribution in this game is not market size; it tracks *liberalization headroom*, because a high-tariff economy brings a large stock of feasible cuts and a low-tariff economy is mechanically near-dummy whatever its GDP. But the near-zero correlation is a property of the *raw* shares, which Table B.2 shows are dominated by each country's own unilateral value. On the cooperative shares the large, import-diversified United States economy comes first, so genuinely cooperative surplus tracks size *more* than headroom, and the "headroom not size" reading reverses once unilateral gains are netted out. What the raw result does establish, and all it establishes, is that the imputation is not a mechanical GDP relabel.

B.4. Is it a relabeled tariff-revenue share?

Once size is rejected, the natural second benchmark is base-year tariff revenue, $R_i = \sum_{n,j} \pi_{ni}^{j} \tau_{ni}^{j,0}$, which is a closer mechanical analogue of marginal contribution under the scalar lever: a country with no base tariff has no surplus to give.

Table B.4: Raw Shapley share against base-year tariff-revenue share, second economy.

N	Pearson	Spearman
3	+0.286	+0.500
4	+0.874	+1.000
5	+0.837	+0.700
6	+0.847	+0.771
7	+0.857	+0.750

The tariff-revenue share of country i is $R_i / \sum_{k \in S_N} R_k$. Source: the benchmark comparison of the replication package.

Three observations. First, the correlation rises sharply from +0.29 at $N = 3$ to 0.84–0.87 at $N \geq 4$: once high-tariff members enter, tariff revenue is the dominant cross-country regularity, and the $N = 3$ figure must not be read into the $N \geq 4$ band. Second, the rankings still diverge: at $N = 7$ the imputation ranks India above the United States and tariff revenue does not. Third, the per-country gap between the two shares reaches +39.8 percentage points (China, $N = 4$) and −25.1 (United States, $N = 3$), with elastic-import economies receiving the largest positive gap.

Interpretation. The imputation is neither a GDP proxy nor a tariff-revenue proxy, but it is much more strongly correlated with the latter, and that correlation is itself N-dependent — weak on the three-member coalition and strong only once a high-tariff member enters. As Section 5.1 states, neither this panel nor the primary one supports a claim that the Shapley structure does something a simple tariff-revenue weighting could not.

B.5. Supermodularity and core membership

A standard sufficient condition for the Shapley value to lie in the core is convexity (Shapley 1971): for every player i and every $S \subseteq T \subseteq N \setminus \{i\}$,

$$v(T \cup \{i\}) - v(T) \;\geq\; v(S \cup \{i\}) - v(S).$$

We test it directly on the coalitional values at every N, enumerating every valid triple.

The violations at $N = 6, 7$ are small — marginal-contribution gaps of \$0.016 to \$0.046 billion, one to two orders of magnitude below the coalitions' own values and within a few multiples of the model's cross-configuration sensitivity band. This is a registered failure of the non-convexity half of a registered conjunctive

Table B.5: Supermodularity violations, second economy.

N	Triples tested	Pass	Violations	Convex to numerical resolution?
3	27	27	0	yes
4	108	108	0	yes
5	405	405	0	yes
6	1458	1449	9 (0.6%)	near
7	5103	5072	31 (0.6%)	near
Pooled $N = 3$–7: $40/7101 = 0.56\%$				

Triples are the valid (i, S, T) with $S \subseteq T \subseteq N \setminus \{i\}$. Source: the monotonicity and supermodularity sweep of the replication package.

claim (threshold: a violation rate of at least 10%; measured, 0.56% pooled), reported as a failure rather than reframed; the companion single-player monotonicity half passes on all 3,019 tested additions (Section C). Tariff complementarity, not strategic substitutability, dominates at every tested N.

Implication. The game is convex to within numerical resolution rather than non-convex, and core membership is confirmed by direct check rather than left as an inference: all 126 proper-coalition constraints hold at $N = 7$, with a minimum slack of 0.37% of $v(N)$, binding at a six-country coalition. This matters for the same reason it matters on the primary panel. Absent the direct check, and given the residual 0.6% of violated triples, coalition stability here would rest on an approximate-convexity argument rather than on a verified fact. As on the primary panel, the check is of the transferable-utility core under the passive-outsider convention; the γ-response convention (Section 2.4) is untested on this panel and is the same open extension recorded in Section 6.

B.6. Validation of the Monte Carlo sampler

Exact enumeration becomes infeasible beyond $N = 7$, so for larger player sets we implement the random-permutation sampler of Castro, Gomez, and Tejada (2009). We validate it against exact enumeration on a three-player economy of the same structure: with $M \in \{50, 200, 1000, 5000\}$ permutations and three independent seeds, the Monte Carlo estimator is within 4σ of the exact Shapley value for every player at every M, and its standard error scales as $1/\sqrt{M}$ to within 5% of the theoretical rate. The validation is methodology-level: it checks the sampler against exact enumeration, not any particular table of coalition values. Every number reported in this paper comes from exact enumeration, so the sampler is not on the path of any result; it is documented here because it is part of the replication package.

B.7. Threat-point sensitivity

The disagreement point in both economies is the base-year applied tariff schedule, which is itself an outcome of prior GATT bargaining and hence endogenous to the object we study (Section 6). We ran a sensitivity check on the three-member coalition replacing it with two synthetic uniform-tariff threat points, one low and one high. The exercise was computed under an earlier characteristic function and has not been recomputed under the definition used in this paper, so we do not report its magnitudes and none of them appears anywhere in this paper. We record the qualitative finding, which does not depend on the levels and which we regard as a limitation rather than a robustness result: the ranking of the allocation is *not* invariant to the threat point, and the total surplus varied over several orders of magnitude across the three scenarios. Threat-point endogeneity is a first-order identification problem for benchmarks of this kind, not a second-order robustness issue. A recomputation under the current definition is one of the extensions in Section 6.

Appendix C. Registered Claims and Their Disposition

Every empirical claim in this paper was registered, with a threshold or verdict band fixed in advance, before the computation that tests it. The tables below list the registered claims that bear on results reported in this paper, together with their disposition against the registered criterion and the section in which each is discussed. Dispositions are reported as recorded, including those that failed, that were downgraded after a placebo, and one that was retracted outright. The identifiers in the first column are those of the pre-registration record, so that each row can be matched to the entry that fixed its criterion in advance.

Claims registered against objects that no longer appear in the paper — superseded specifications, and two claims orphaned when the analysis was re-centered on the primary panel — are omitted here; they carry no live claim, and the pre-registration record lists them in full.

Table C.1: Registered claims: the primary panel.

ID	Registered claim	Disposition	Reported
P-07″	US fairness gap clears +5pp	**FAIL** — +3.29pp on the primary branch; clears only on the alternative branch	Section 4.6
P-08	The Shapley-consistent weight set is a corner of the weight box	**INCONCLUSIVE** — the search never reached differentiability	Section 4.3
P-09	Test of the implied weight against BSY's estimate	**INCONCLUSIVE** — same cause	Section 4.3
P-17	The benchmark's tariff vector fits BSY's 151 negotiated moments	**PASS** on the literal criterion; cite only with the placebo-adverse rider	Section 4.5
P-18	An untruncated policy space still clears the necessary condition	**PASS** — reported as a placebo, never as a specification	Section 5.2
P-19	The grand-coalition optimum is interior at every member	**STRONG-FORM FAIL** — South Korea at the boundary; the weak form (five of six) holds *post hoc*	Section 4.6
P-20	The US outsider gain is negative on a majority of coalitions excluding it	**FAIL** — negative on 13 of 31	Section 4.6
P-23	Core membership, all 62 proper coalitions	**PASS** — transferable-utility core, passive-outsider convention	Section 4.2
P-24	Three-term decomposition of the dollar identity	**DESCRIPTIVE** — identity verified to 1.4×10^{-17}	Section 4.2
P-26	Implementation requires a transfer of $\approx$ 19% *to* the US	**RETRACTED** — sign inverted; the US pays and Australia is the loser	Section 4.6
P-27	Sensitivity of the fairness gap across four allocation rules	**DESCRIPTIVE** — in-sample, $n = 6$, no significance statement	Section 4.3
P-35	Violation count and signature of single-player monotonicity	**SPLIT** — count reproduces exactly; the signature is neither necessary nor sufficient, and Canada appears in 16 of 17	Section 4.2

Table C.2: Registered claims: the efficiency decomposition, the external comparison, and the second economy.

ID	Registered claim	Disposition	Reported
P-29, P-31	The line-by-line negotiated-only frontier	**REFUTES** the reversal reading; $F^{v2} = 0.4368$, and the naive gap mixes coverage with instrument granularity	Section 4.4
P-33	Is the within-agenda shortfall a transferability artifact?	**RESOLVED** — no: the constraint costs 1.2% of $v(N)$ and the shortfall is 16.06%	Section 4.4
P-34	The full-coverage line-by-line frontier	**DESCRIPTIVE** — value certified across starts, argmax not; quoted as a value only	Section 4.4
P-36	A no-transfer improvement over BSY's realized outcome exists	**FALSIFIES** the strong form — the improvement is only *weak*: four of six members are exactly indifferent	Section 4.4
P-38	"Widening the agenda is worth about four times bargaining harder"	**DESCRIPTIVE** — 3.88×, a point estimate on both operands only once both are floor- and regime-matched	Section 4.4
P-39	Is BSY's schedule a feasible point of our cuts-only box?	**FAILS**, but by far less than first reported: 5 of 151 lines, all negative; an earlier reading of 86 increases was solver tolerance and is retracted	Section 4.4
P-40, P-41, P-43	Separating cut depth from the bilateral architecture	**NO VERDICT BANKED** — boundary result; the reconstruction stalled with no estimate; the existence hypothesis is not proven	Section 4.4
P-37	The benchmark's cuts track observed 1990–2000 liberalization, $\rho \geq 0.3$	**PASS** on the literal band, **adverse** to its own placebo; no significance language attaches	Section 4.5
P-42, P-45	Head-to-head against BSY's own fitted schedule	**NOT RESOLVED** at the terminal caliber — resolved on the pooled sample, not on the coverage-matched one	Section 4.5
P-10	Passive-outsider gain positive on a majority of coalitions	**PASS**	Section B
P-11	Preconditions for share language, $v(N) > 0$ and $\phi_i(v) \geq v(\{i\})$	**PASS** on both economies	Section 4.2, Section B
P-12	Supermodularity fails on at least 10% of triples, and monotonicity fails	**FAIL** of the registered conjunctive form — 0.56% pooled, and monotonicity does not fail here	Section B.5
P-13	Low-tariff members are near-dummy	**PASS**	Section B
P-16	The US is the largest cooperative contributor	**PASS** at every $N \geq 4$	Section 5.1
P-22	Second-economy horse race against the revenue share	**INCONCLUSIVE** — on the primary panel the result is driven entirely by Japan, and dropping it moves every leg adverse	Section 5.1
P-28	Core membership, all 126 proper coalitions	**PASS**	Section B.5

Appendix D. Replication Package

The package contains the programs, the calibration inputs that may be redistributed, and one machine-readable result file for every table and figure in this paper. Each result file records the program and the options that produced it, the version of the code base, the random seed, the software environment, the wall-clock runtime, and a timestamp, so that any number in the paper can be traced to the run that generated it.

Primary panel: replication and application. The reading and fidelity check of the calibrated primitives (Section 3.1); the hat-algebra equilibrium system and the Nash-in-Nash bargaining problem; the cold-start parity checks behind the equilibrium-map replication; the weight re-estimation attempt reported as a negative result (Section 4.1); the replication of the published counterfactuals; the 63-coalition solve, its invariant checks and the dollar-identity decomposition (Section 4.2); and the sensitivity, placebo and pre-registered sanity checks of Sections 4.3 and 5.2.

Line-by-line programs. The four nested optimizations behind Table 5 — F^{v2}, F^{IR}, D, D^{IR} — the Pareto-dominant frontier, the bilateral-pair reconstruction, and the rank comparisons against observed tariff changes (Sections 4.4 and 4.5), each run from multiple starts with per-start dispersion recorded.

Second economy. The 31-region $\times$ 40-sector calibration; the hat-algebra equilibrium system; a replication of the source paper's own published tables; the coalitional value function; the exact Shapley enumeration for $N = 3$ to 7, with coalition values cached across N; and the Monte Carlo sampler with its validation (Section B.6).

Reproducibility. The cost of reproducing the primary panel end to end is dominated by the 63-coalition solve and by the four line-by-line optimizations. Reproducing the second economy's exact enumeration for $N = 3$ to 7 takes approximately five to six hours on a single processor: the cache is keyed on the coalition rather than on the player set, so a coalition already priced at one N is never re-solved at a larger one.